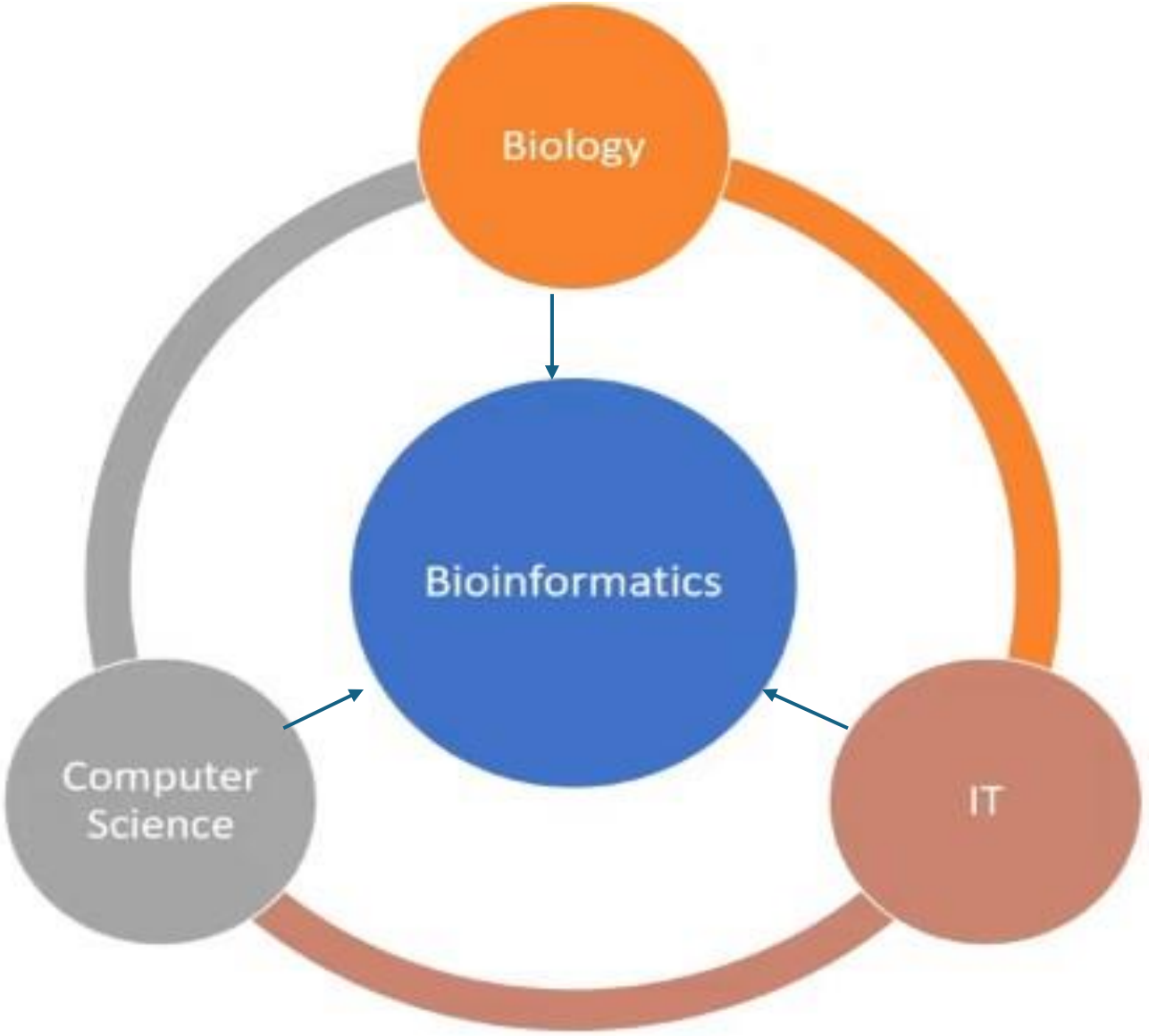
What is the Course about?

- Identify the basic principles of bioinformatics and computational biology.
- Gain knowledge of bioinformatic tools on the internet.
- Know new concepts such as phylogenesis, protein structure prediction, and comparative genomics.

What is Bioinformatics?

Merge and emerged as a single discipline is called **Bioinformatics**



Bioinformatics Definition



Store



Retrieve



Analyze

What is Bioinformatics ?

"the science helpful to store, retrieve and analyze the large amount of biological information".

Applications of Bioinformatics

- Data analysis using machine learning and AI
- Big data management in genomic databases
- Algorithm design for sequence alignment and structure prediction
- Development of bioinformatics software and tools
- Computational genomics and personalized medicine
- Visualization of complex biological information

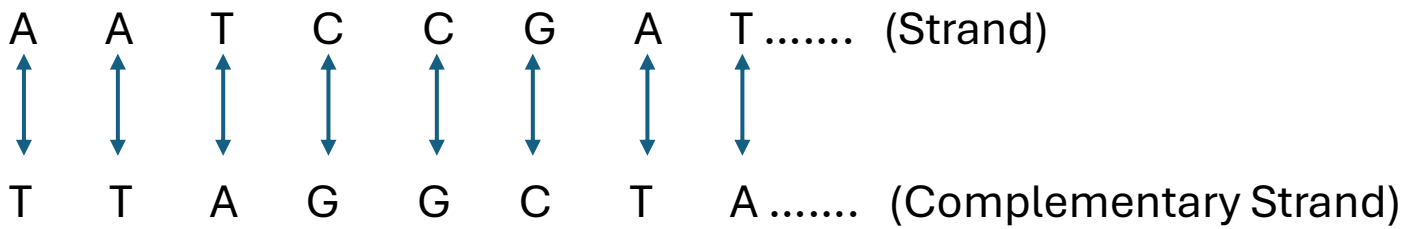
Key Areas of Bioinformatics Work

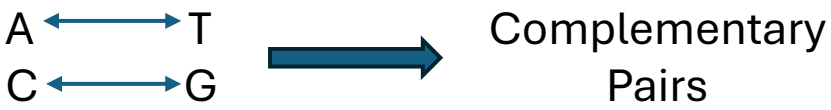
- DNA
- RNA
- Protein
- Phylogeny
- Computational Biology
- Machine learning in Bioinformatics

DNA

(Deoxyribonucleic acid)

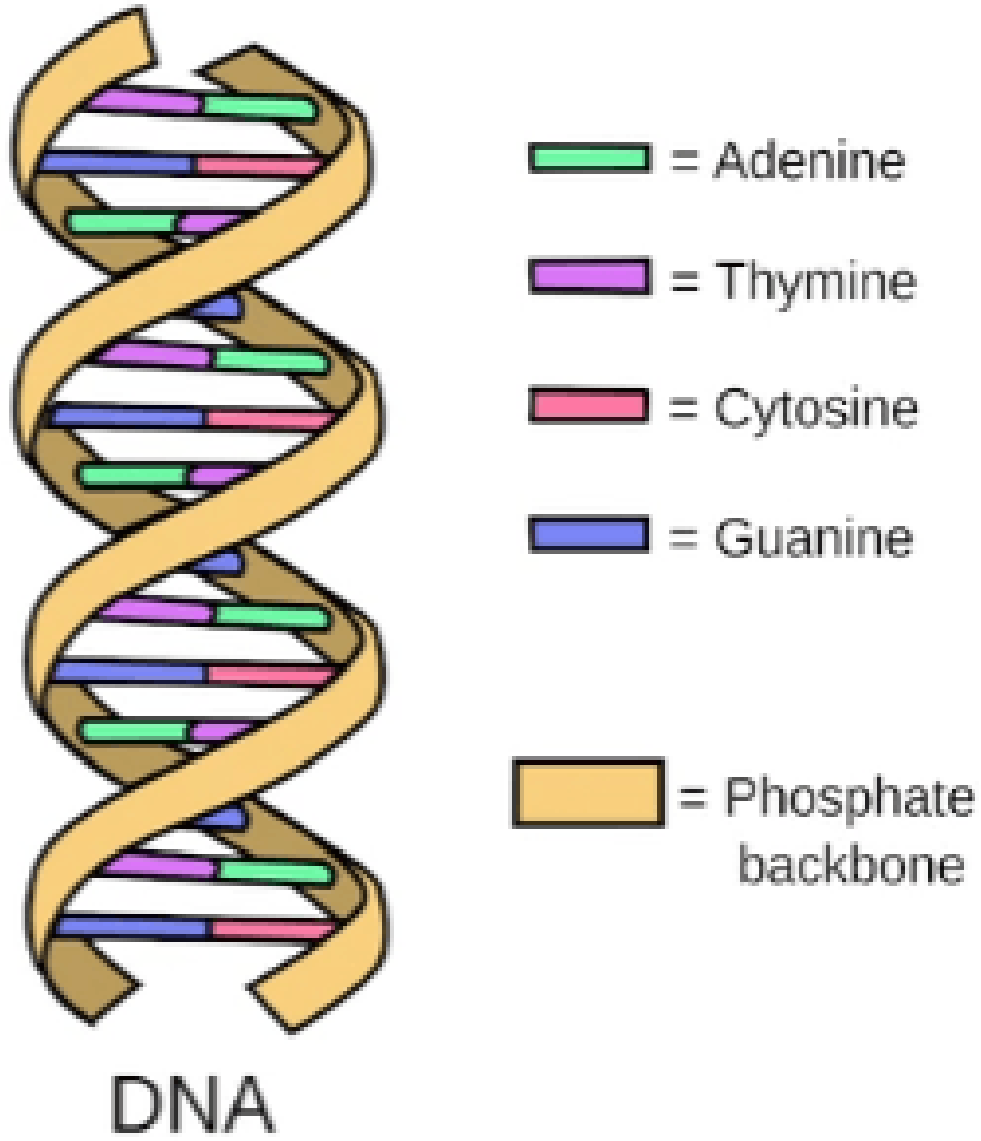
- DNA is a string of full A(adenine), T(thymine), C(cytosine), G(guanine).





- **Deoxyribonucleic acid**
- Contains **Genetics** Instruction
- Long Sequence of **4 types Nucleotides/bases**:
  - Adenine
  - Thymine
  - Cytosine
  - Guanine

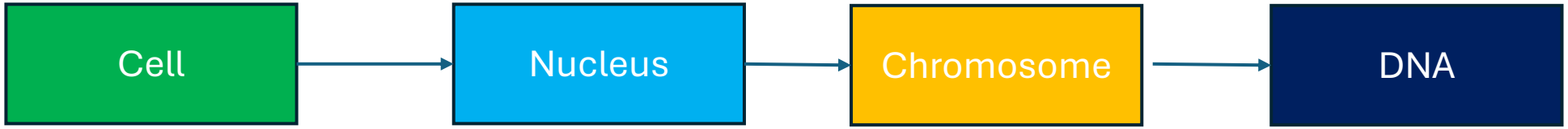
- In 1953, James Watson and Francis Crick discerned the **double helix structure** of DNA.



DNA

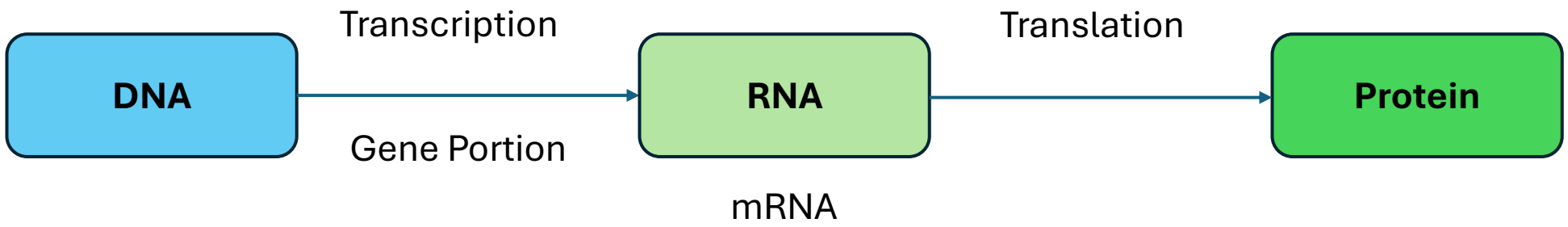
3 billions pair in human Genome/DNA.

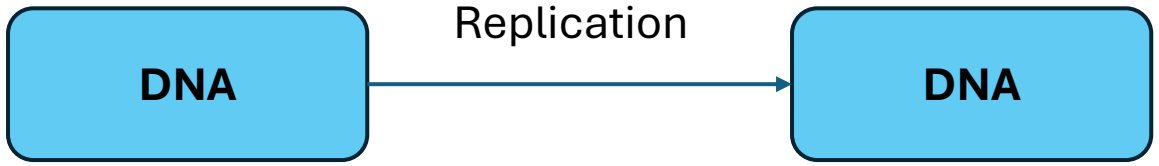
Where is DNA?



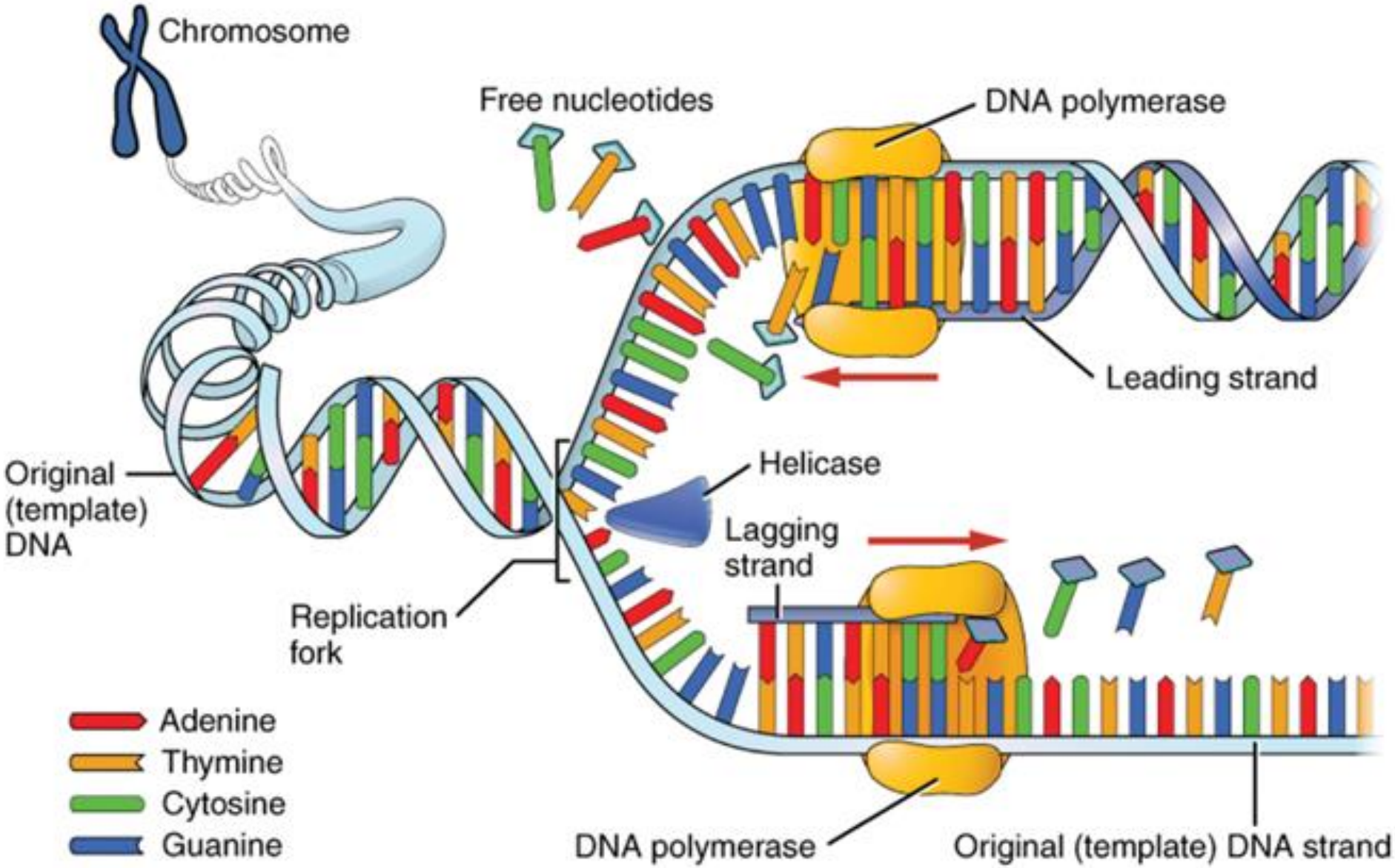
- ❑ **Expression of DNA is Gene.**
- ❑ **A small portion of DNA is Gene.**

Central Dogma

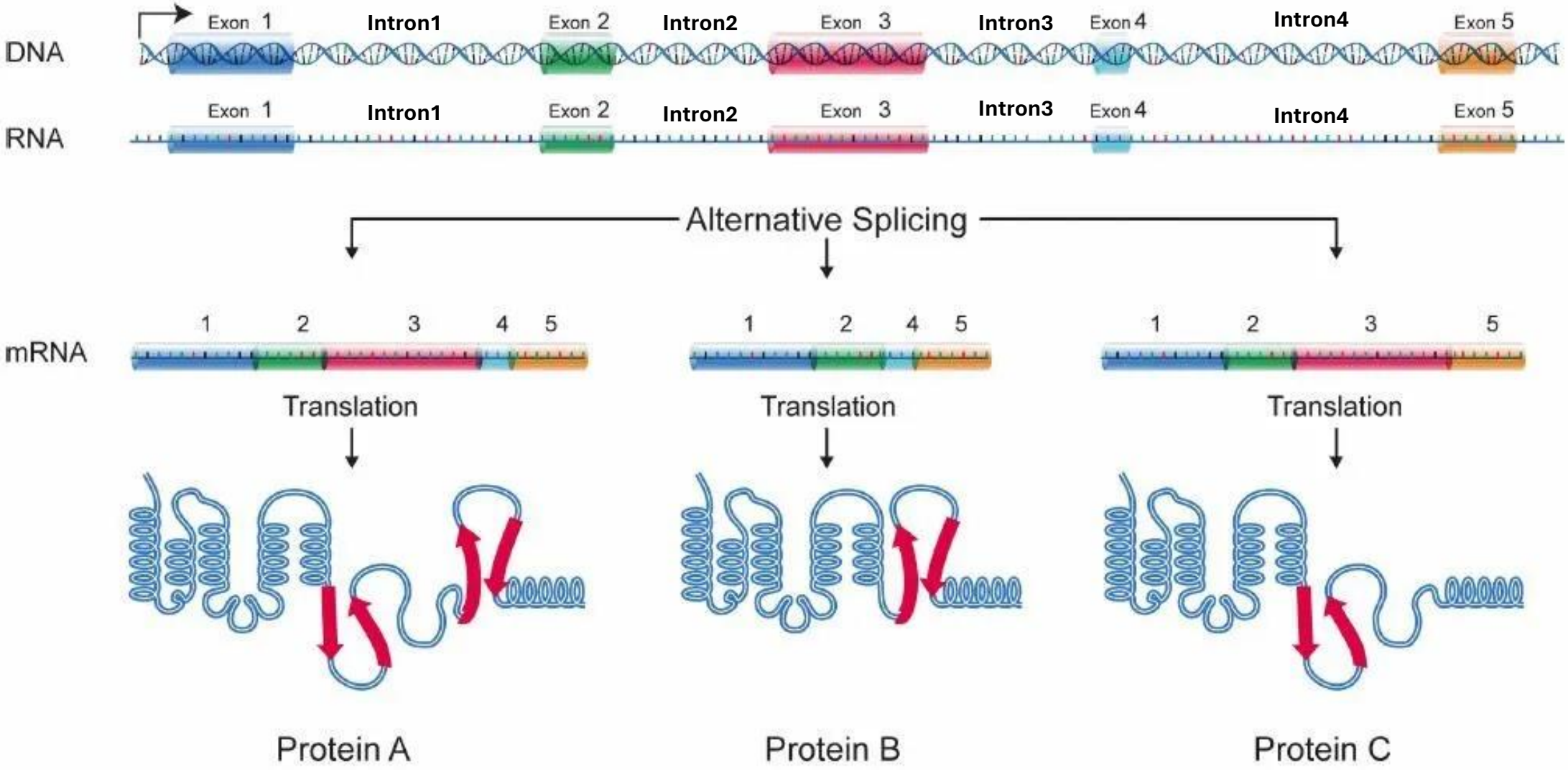




DNA ⟶ DNA


Chromosome
Free nucleotides
DNA polymerase
Leading strand
Original (template) DNA
Helicase
Lagging strand
Replication fork
Adenine
Thymine
Cytosine
Guanine
DNA polymerase
Original (template) DNA strand

Eukaryotic gene structure



RNA

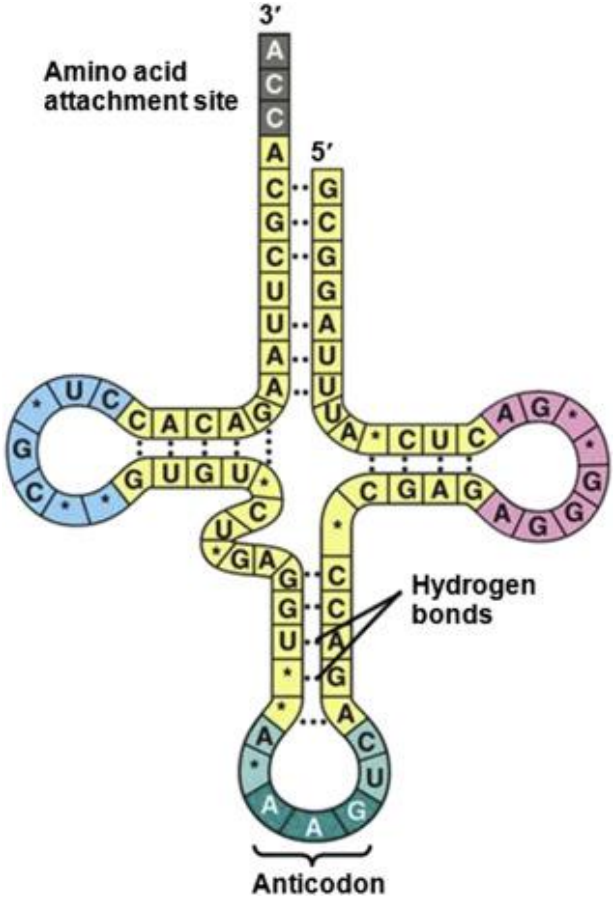
RNA (Ribonucleic Acid) is a nucleic acid molecule that plays several essential roles in coding, decoding, regulation, and expression of genes. It is similar to DNA but differs in a few key ways.

- **In DNA**, the four nitrogen bases are:
  Adenine (A), Thymine (T), Cytosine (C), Guanine (G).

- **In RNA**, **Thymine (T)** is **replaced by Uracil (U)**.
  So, the bases in RNA are:
  Adenine (A), Uracil (U), Cytosine (C), Guanine (G).
- RNA is single Stranded.

Basic Pairing Rules:

In DNA	In RNA
A ↔ T	A ↔ U
G ↔ C	G ↔ C



Types of RNA and Their Functions

Type	Full Form	Function
mRNA	Messenger RNA	Carries genetic information from DNA to ribosomes for protein synthesis.
tRNA	Transfer RNA	Brings amino acids to the ribosome during protein synthesis.
rRNA	Ribosomal RNA	Forms part of the structure of ribosomes and helps catalyze protein synthesis.

- **mRNA** receives code for Protein Synthesis, **tRNA** transfer this code to **rRNA** , **rRNA** makes protein using this code

Key Differences Between DNA and RNA

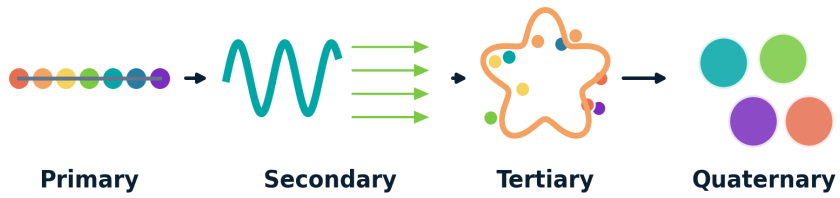
No.	Feature	DNA	RNA
1	**Sugar**	Deoxyribose	Ribose
2	**Nitrogen Base**	Thymine (T) present	Uracil (U) replaces Thymine
3	**Structure**	Double-stranded (Double helix)	Single-stranded
4	**Function**	Stores genetic information	Helps in protein synthesis (mRNA, tRNA, rRNA)
5	**Location**	Found mainly in nucleus	Found in nucleus and cytoplasm

Proteins, Codons & Protein Structure

What You Will Learn

A clear roadmap for today's lecture

- Define protein, amino acid, polypeptide and codon in simple terms.
- Explain how RNA codons map to amino acids during translation.
- Calculate why 4 RNA bases can create 64 possible codons.
- Compare primary, secondary, tertiary and quaternary protein structures.
- Connect protein structure knowledge with bioinformatics databases and tools.



→

Lecture focus

From biological sequence → molecular structure → bioinformatics interpretation.

Why Proteins Matter in Bioinformatics

Proteins are biological machines encoded by genetic information

F

Function

Enzymes, receptors, antibodies, transporters and structural molecules.

D

Disease

Mutations can change amino acid sequence, folding and function.

Rx

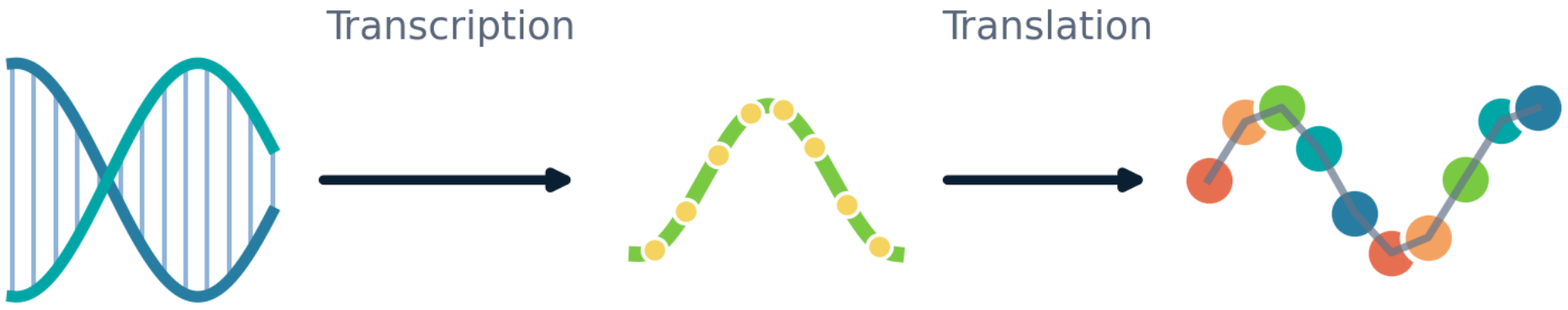
Drug design

Many medicines target protein active sites or protein interactions.

- Bioinformatics helps us store, compare, predict and interpret protein sequences and structures.
- Protein sequence data can be used for homology search, function prediction and evolutionary analysis.
- Protein structure data can help explain how a protein works in 3D space.

From Genetic Information to Protein

Bioinformatics often connects DNA/RNA sequence data with protein-level meaning



DNA stores information

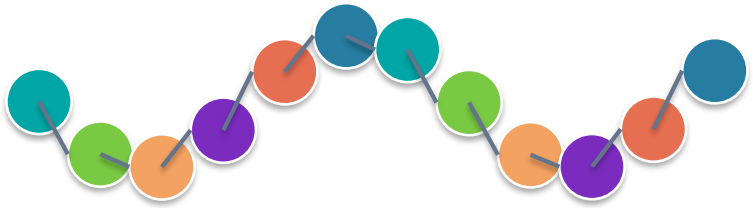
RNA carries the message

Protein performs function

Protein: Basic Definition

The lecture begins from the idea that protein is a chain of amino acids

- A protein is a biological macromolecule made from amino acids.
- A chain of amino acids is called a polypeptide.
- Proteins fold into specific structures to perform specific functions.
- The human body and all living cells depend on proteins for life processes.



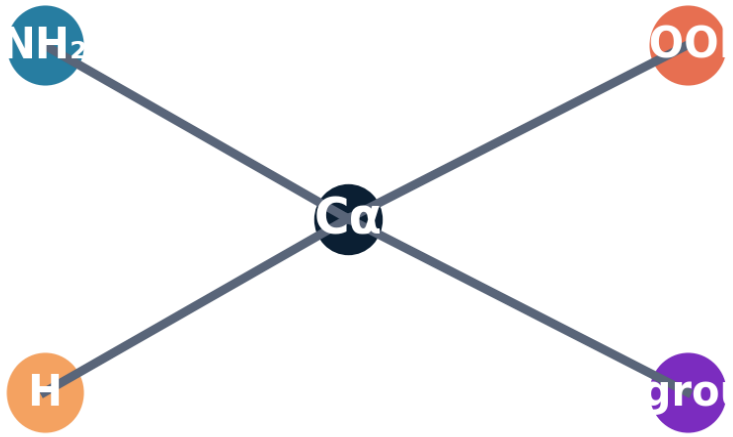
Key idea

Sequence decides the folding pathway and final function.

Amino Acids: Building Blocks

Every amino acid has a common backbone and a variable side chain

General amino acid structure



- Amino group: —NH_2
- Carboxyl group: —COOH
- Hydrogen atom: —H
- R group: variable side chain that gives each amino acid its unique property.

In proteins, the R group strongly affects folding, charge, polarity and interaction with water.

20 Standard Amino Acids

The standard genetic code uses 20 common amino acids to build proteins

Class	Examples	Simple property
Nonpolar	Gly, Ala, Val, Leu, Ile, Met, Pro	Hydrophobic tendency
Aromatic	Phe, Tyr, Trp	Ring structures
Polar uncharged	Ser, Thr, Cys, Asn, Gln	Can form H-bonds
Acidic	Asp, Glu	Negative charge
Basic	Lys, Arg, His	Positive charge

A

One-letter codes

Examples: A = Alanine, G = Glycine, M = Methionine, W = Tryptophan.

Aa

Three-letter codes

Examples: Ala, Gly, Met, Trp. These appear in many databases and textbooks.

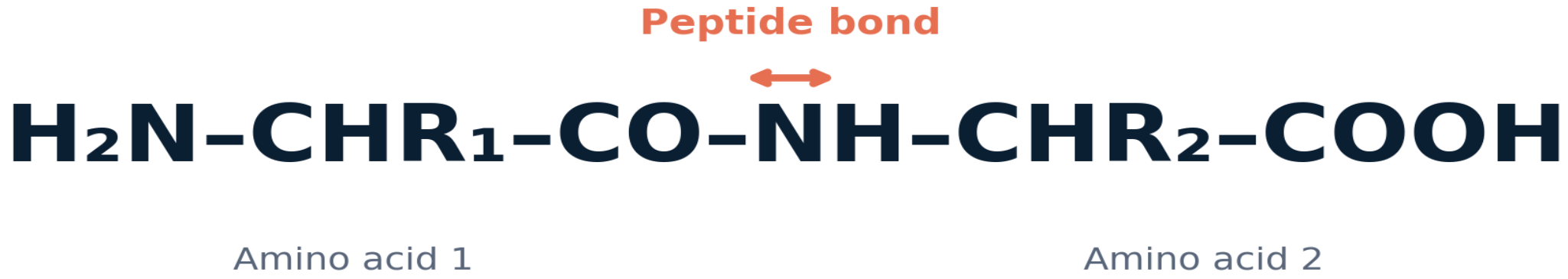
R

Side-chain logic

Side-chain chemistry influences 3D folding and biological activity.

Peptide Bond & Polypeptide Chain

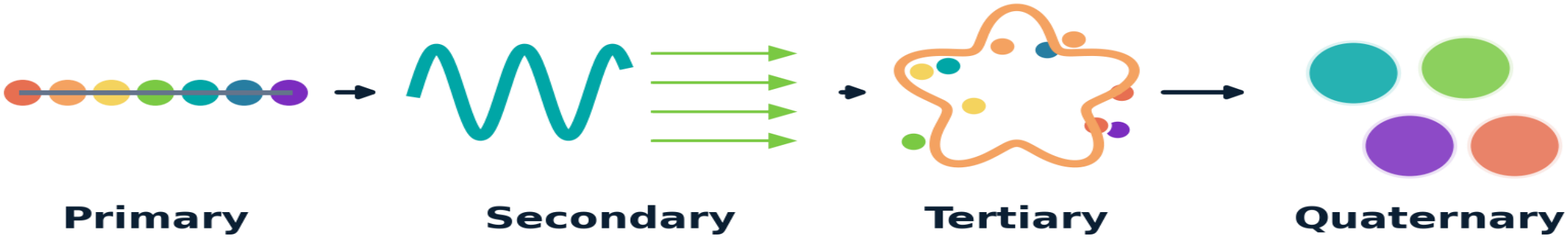
Amino acids are joined by peptide bonds



- A peptide bond links the carboxyl group of one amino acid to the amino group of another.
- This creates a repeating protein backbone.
- The chain has direction: N-terminus → C-terminus.
- The order of amino acids forms the primary structure.

Four Levels of Protein Structure

Protein structure is hierarchical



Primary

linear amino acid sequence

Secondary

local alpha helix / beta sheet

Tertiary

3D shape of one chain

Quaternary

multiple chains together

SECTION 1

Primary Structure

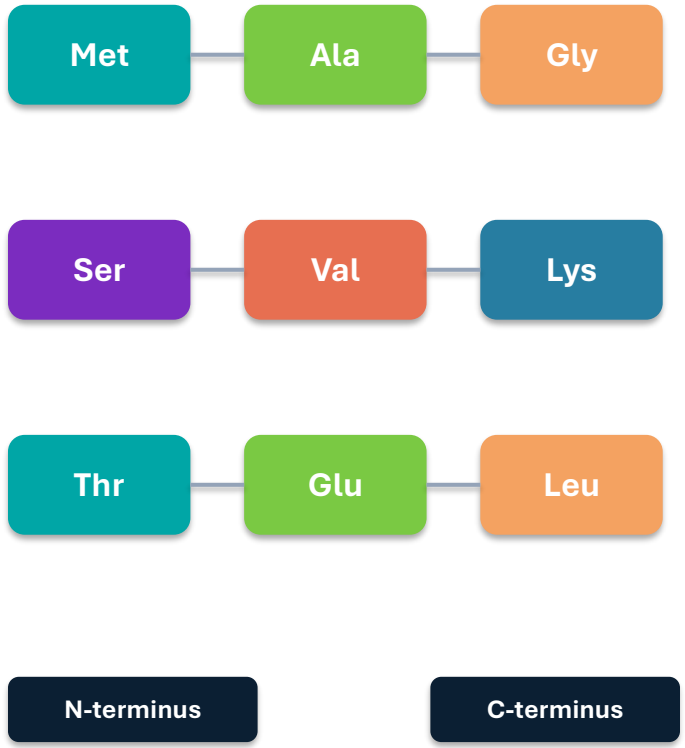
The exact order of amino acids in a polypeptide chain



Primary Structure

The first level of protein structure

- Primary structure is the linear sequence of amino acids in a polypeptide chain.
- Amino acids are joined by peptide bonds.
- Even the same amino acids in a different order can produce a different protein.
- Primary sequence is the foundation for all higher-level folding.



Reading a Protein Sequence

Bioinformatics databases usually store protein sequences as strings

Example sequence

M A G S V K T E L

- Sequences are commonly written from N-terminus to C-terminus.
- One-letter amino acid codes make long sequences compact.
- A sequence can be saved in FASTA format for database search and analysis.
- Protein sequence comparison helps identify similarity, function and evolution.

```
>Protein_example
MAGS VKTEL AQLK
GGHILKPNVTS
...
```

FASTA is a common plain-text format used for sequence data.

RNA, Nucleotides & Codons

Protein sequence is read from RNA in groups of three bases

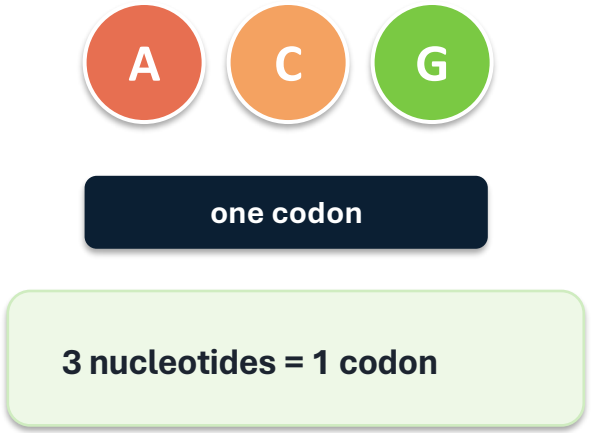
A
Adenine

C
Cytosine

G
Guanine

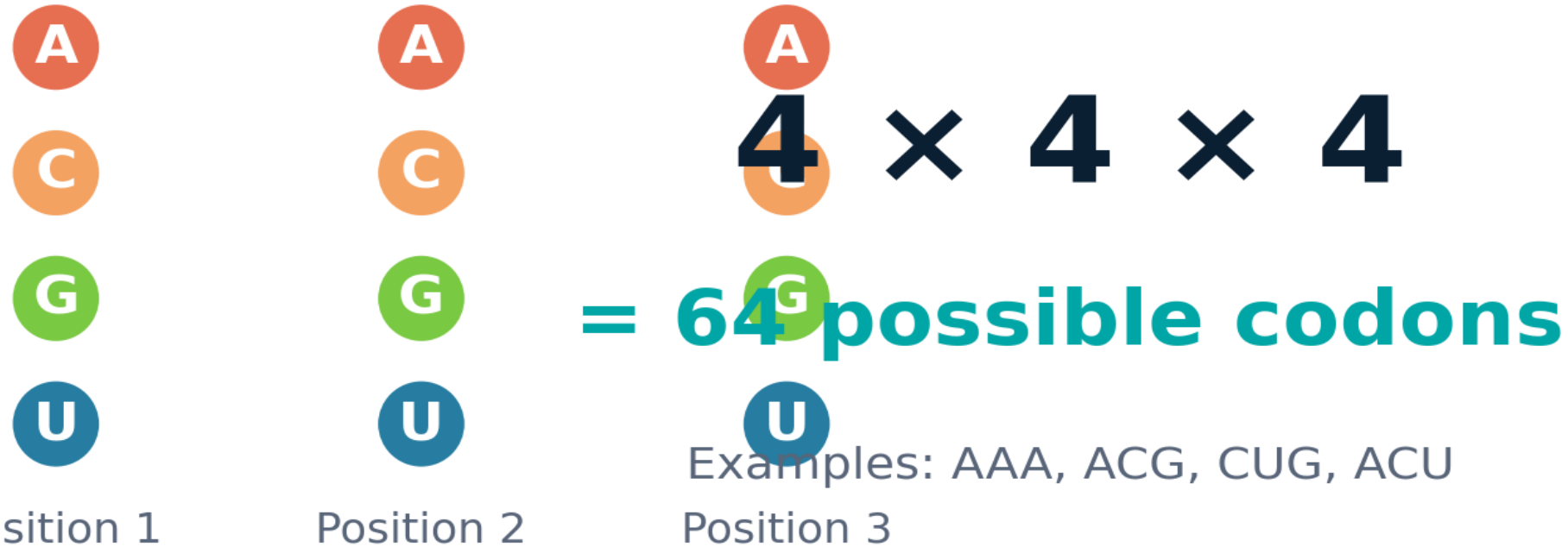
U
Uracil

- RNA has four nucleotides: A, C, G and U.
- A codon is a group of three RNA nucleotides.
- During translation, codons guide which amino acid is added next.
- Example codons: AAA, ACG, CUG, ACU.



Why 4 × 4 × 4 = 64 Codons

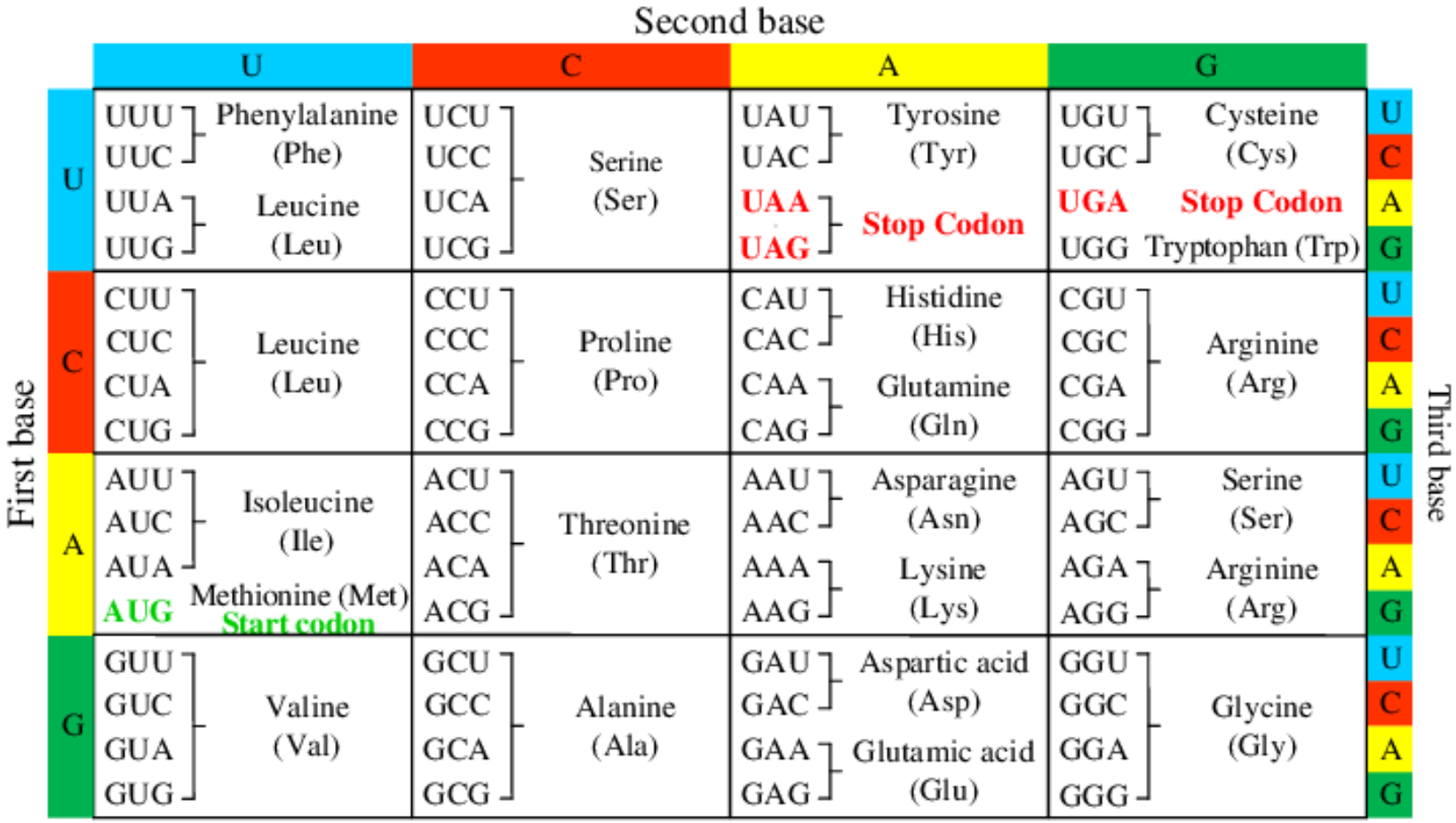
Three positions, each with four possible RNA bases



Because each of the 3 positions can be A, C, G or U, total possibilities = 4^3 = 64.

Genetic Code Mapping

The standard genetic code translates RNA codons into amino acids or stop signals



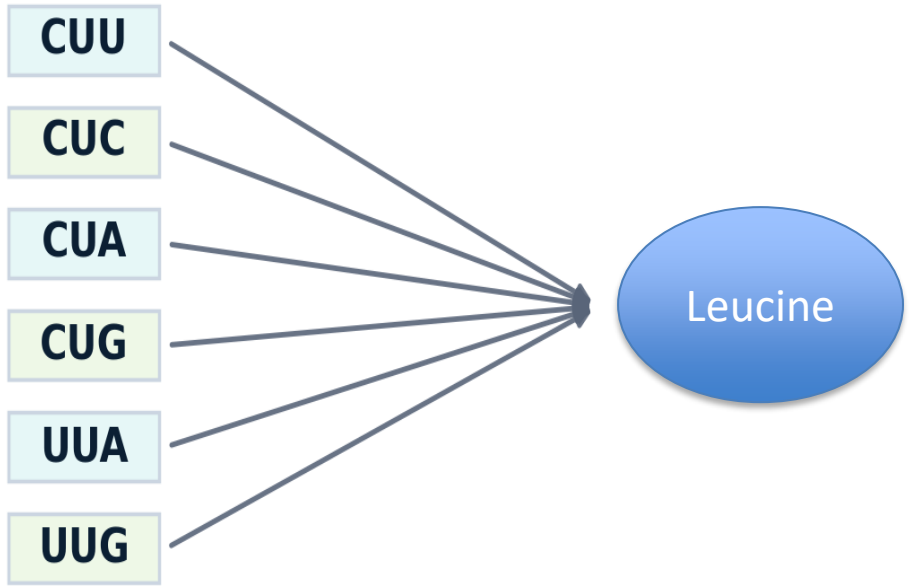
- 64 possible codons exist.
- Most codons specify amino acids.
- Some codons act as stop signals.
- AUG usually codes Methionine and often works as the start codon.

> **Important**
>
> Genetic code is many-to-one, not one-to-one.

Many-to-One Mapping: Codon Degeneracy

Several codons may encode the same amino acid

Different codons can code for the same amino acid



This is called genetic code degeneracy

- There are 64 codons but only 20 standard amino acids.
- Therefore, many amino acids are encoded by more than one codon.
- This redundancy is called degeneracy of the genetic code.
- It can reduce the effect of some mutations because different codons may still code the same amino acid.

Start & Stop Codons

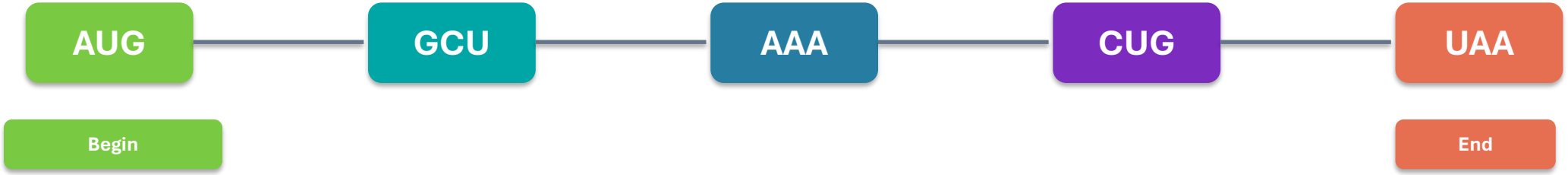
Translation needs signals for beginning and ending

Start codon

AUG commonly starts translation and codes for Methionine.

Stop codons

UAA, UAG and UGA signal termination of translation.



SECTION 2

Secondary Structure

Local folding patterns: α-helix and β-pleated sheet



Secondary Structure

Local folding of the protein backbone

- Secondary structure means local folding of the polypeptide chain.
- Two common patterns are α-helix and β-pleated sheet.
- These structures are mainly stabilized by hydrogen bonds.
- The hydrogen bonds form between backbone atoms, not usually between side chains.



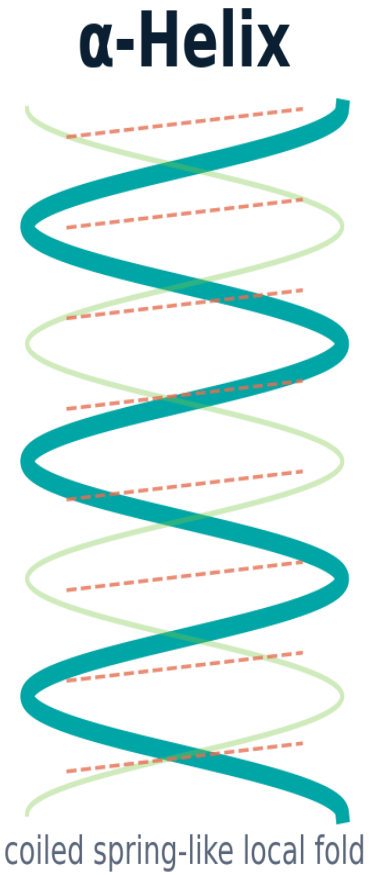
H

Backbone interaction

The repeating backbone allows regular hydrogen-bond patterns.

α-Helix

A coiled spring-like secondary structure

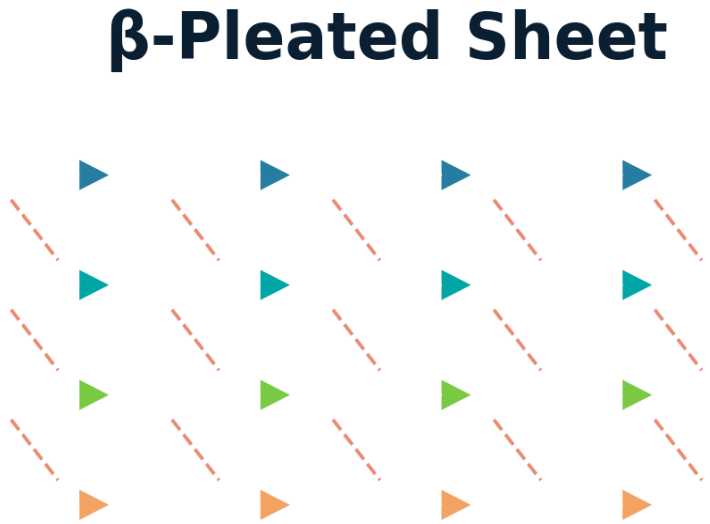


- The chain coils into a helix shape.
- Hydrogen bonds help hold the helix together.
- Side chains extend outward from the helix.
- Alpha helices are common in many proteins, including membrane proteins and enzymes.

Memory clue: α-helix looks like a spring.

β-Pleated Sheet

A folded zigzag pattern formed by beta strands



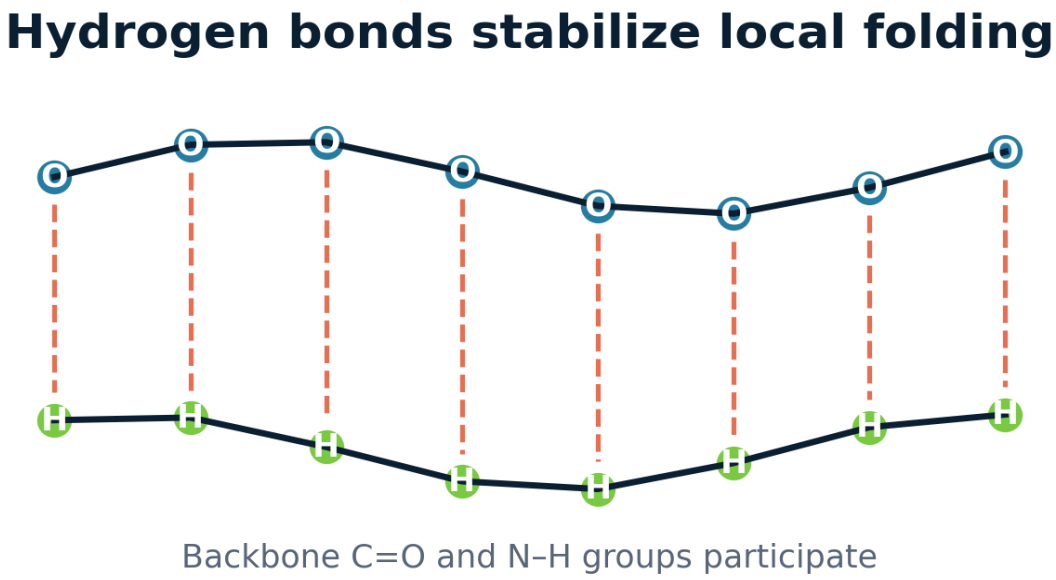
zigzag strands held together by hydrogen bonds

Memory clue: β-sheet looks like folded paper.

- Beta strands align beside each other to form sheets.
- The sheet looks like a folded or pleated surface.
- Hydrogen bonds form between neighboring strands.
- Strands may be parallel or anti-parallel depending on direction.

Hydrogen Bonds in Secondary Structure

The main stabilizing force for α-helices and β-sheets



- Hydrogen bonds are weak individually but powerful collectively.
- They occur between electronegative atoms and hydrogen atoms.
- In secondary structure, they commonly involve backbone C=O and N–H groups.
- This regular bonding creates predictable folding patterns.

Important: secondary structure describes local shapes; the whole 3D shape is tertiary structure.

SECTION 3

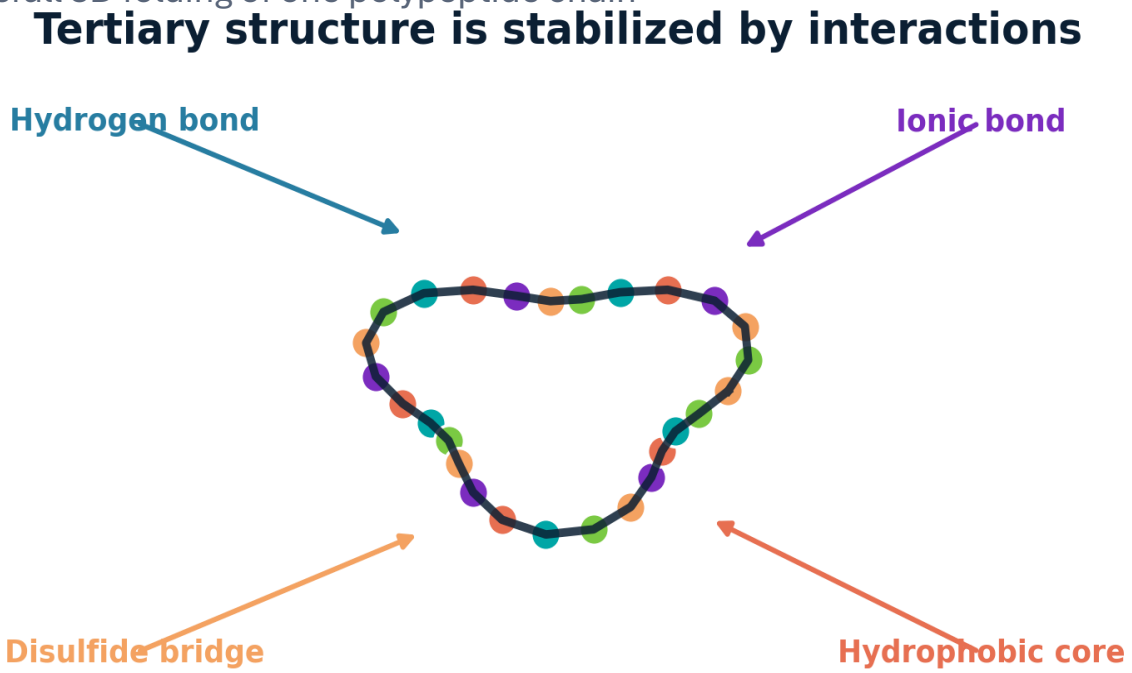
Tertiary Structure

The overall 3D shape of a single polypeptide chain



Tertiary Structure

Overall 3D folding of one polypeptide chain



- Tertiary structure is the complete 3D shape of one polypeptide chain.
- It depends on the sequence and the chemistry of side chains.
- Folding brings distant amino acids close together in 3D space.
- The active site of an enzyme often forms only after tertiary folding.

Primary sequence influences tertiary folding, which strongly affects protein function.

Forces Stabilizing Tertiary Structure

Different chemical interactions hold the 3D shape together

Hydrogen bonds

Weak interactions involving polar atoms.

Ionic bonds

Attractions between oppositely charged side chains.

Disulfide bridges

Covalent bonds between cysteine residues.

Hydrophobic interactions

Nonpolar side chains cluster away from water.

- Correct folding is essential for biological activity.
- Misfolding can reduce function or cause disease.
- Bioinformatics tools can predict domains, motifs and sometimes structures from sequence data.

SECTION 4

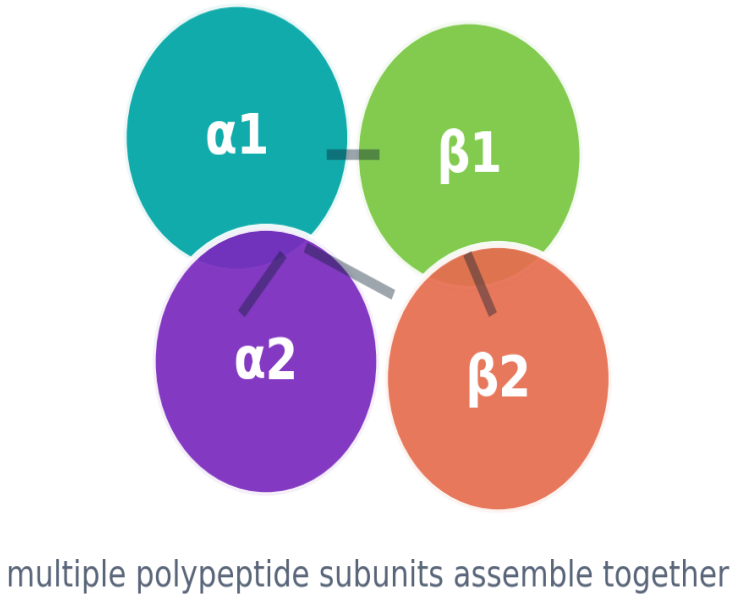
Quaternary Structure

When two or more folded polypeptide chains work together



Quaternary Structure

Assembly of multiple polypeptide subunits

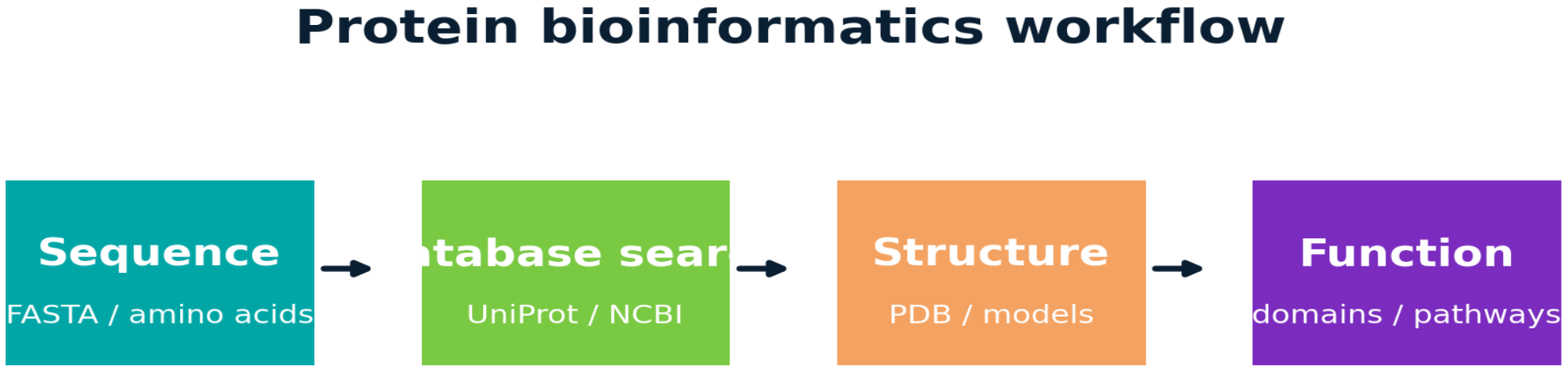


- Quaternary structure forms when two or more polypeptide chains join together.
- Each chain may already have its own tertiary structure.
- Subunits can be identical or different.
- Example: hemoglobin has four subunits that cooperate to carry oxygen.

Quaternary = multi-chain functional complex.

Protein Structure in Bioinformatics

From sequence data to functional interpretation



NCBI / GenBank

nucleotide and protein-related records

UniProt

protein sequence and function information

RCSB PDB

3D structures of biological macromolecules

Class Practice / Quick Check

Use these questions to test understanding



- If RNA has four bases, how many possible codons are possible with three-base codons?
- Why can 64 codons encode only 20 amino acids?
- Which level of protein structure is the linear amino acid sequence?
- What is the difference between tertiary and quaternary structure?
- Which bonds mainly stabilize secondary structure?

Activity

Pick any three codons and identify the amino acids using the codon table.

Key Takeaways

Final summary and Q&A

1

1. Proteins

Chains of amino acids joined by peptide bonds.

2

2. Codons

RNA triplets map genetic information to amino acids.

3

3. Structure

Protein structure has four hierarchical levels.

- Primary: linear sequence.
- Secondary: α-helix and β-sheet.
- Tertiary: 3D shape of one chain.
- Quaternary: multiple chains together.

References / sources:

- User-provided Lecture 2 PPTX (course source).
- NCBI Bookshelf: Primary Protein Structure.
- RCSB PDB-101: Protein hierarchical structure.
- NCBI/PMC: Standard genetic code mapping.
- UniProt: Protein sequence and function resource.

Thank You • Any Questions?

Epigenetics(Post Replication Modification)

What is Epigenetics?

Epigenetics is the study of **how genes are turned on or off without changing the DNA sequence**.

DNA doesn't change, but chemical modifications control whether a gene is active.

Main mechanisms:

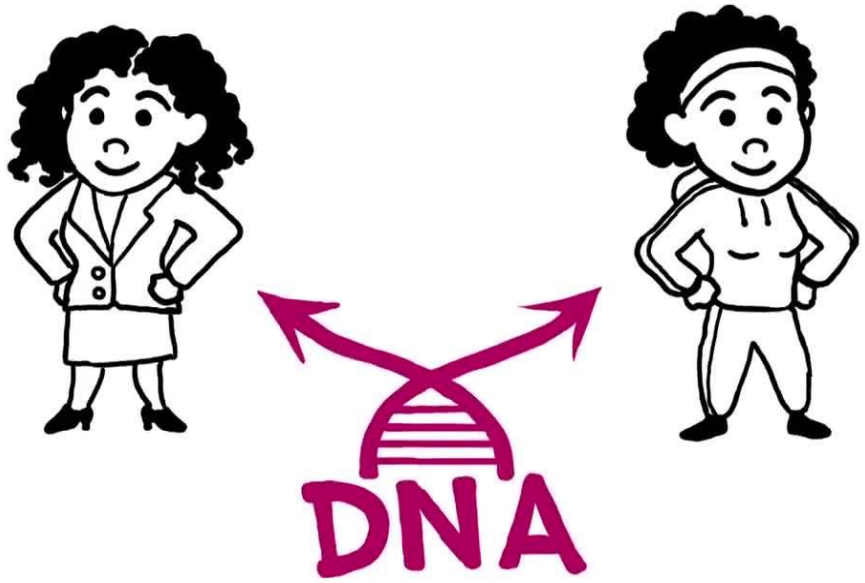
- **DNA methylation:** Adds chemical tags to DNA → usually turns genes off.
- **Histone modification:** Changes how DNA is packed → affects gene activity.
- **Non-coding RNAs:** Small RNAs can block gene expression.

Why it matters: Explains cell differentiation, development, and diseases like cancer.

💡 **Simple idea:** Genes are like a book; epigenetics decides **which pages are open or closed**, without rewriting the book.

Epigenetics

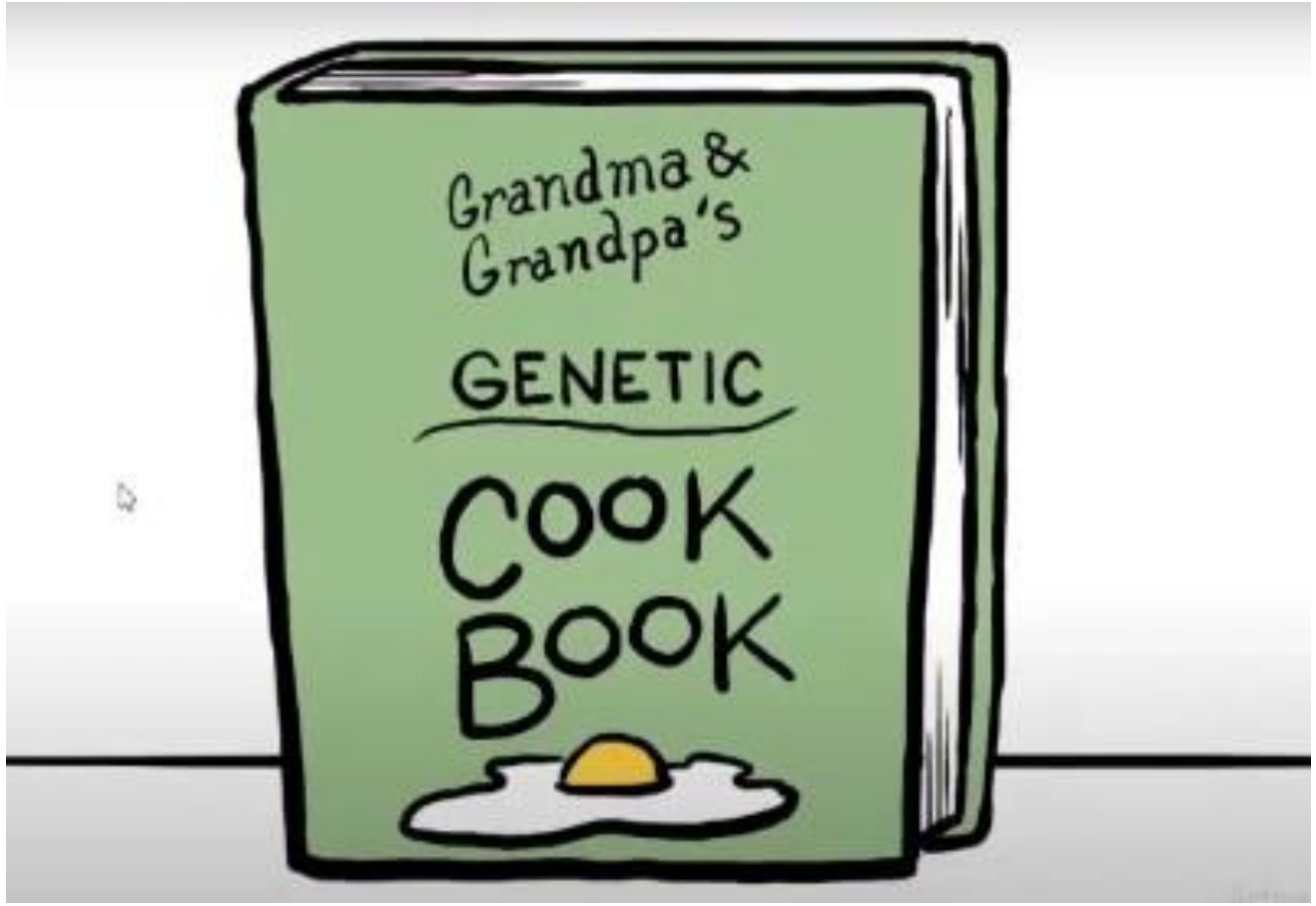
- **Identical twins** have **exactly the same DNA** because they come from the same fertilized egg.
- Despite having the same genes, twins can have **differences in appearance, health, or behavior**.
- This happens because of **epigenetics**: chemical changes (like DNA methylation or histone modification) can turn genes on or off differently in each twin.



Epigenetics

Example with Twins:

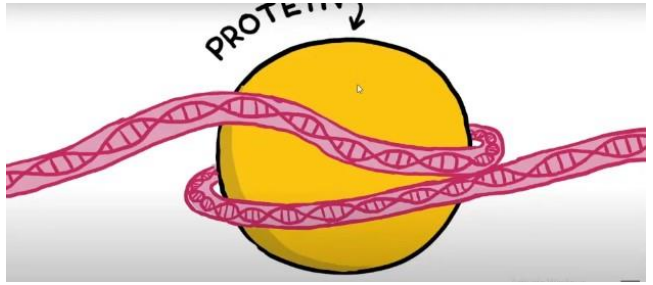
- Identical twins have the same “recipe book.”
- But epigenetic marks act like **personal cooking choices**, so each twin ends up with a slightly different “dish” (traits, health, appearance).



Epigenetics

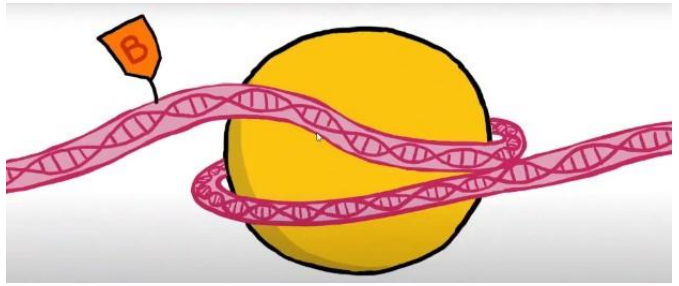
Epigenome







Free radical = “unstable molecule that can harm cells”
CH3, NH2,COOH



Epigenetics



Epigenetics can be passed from one generation to the next

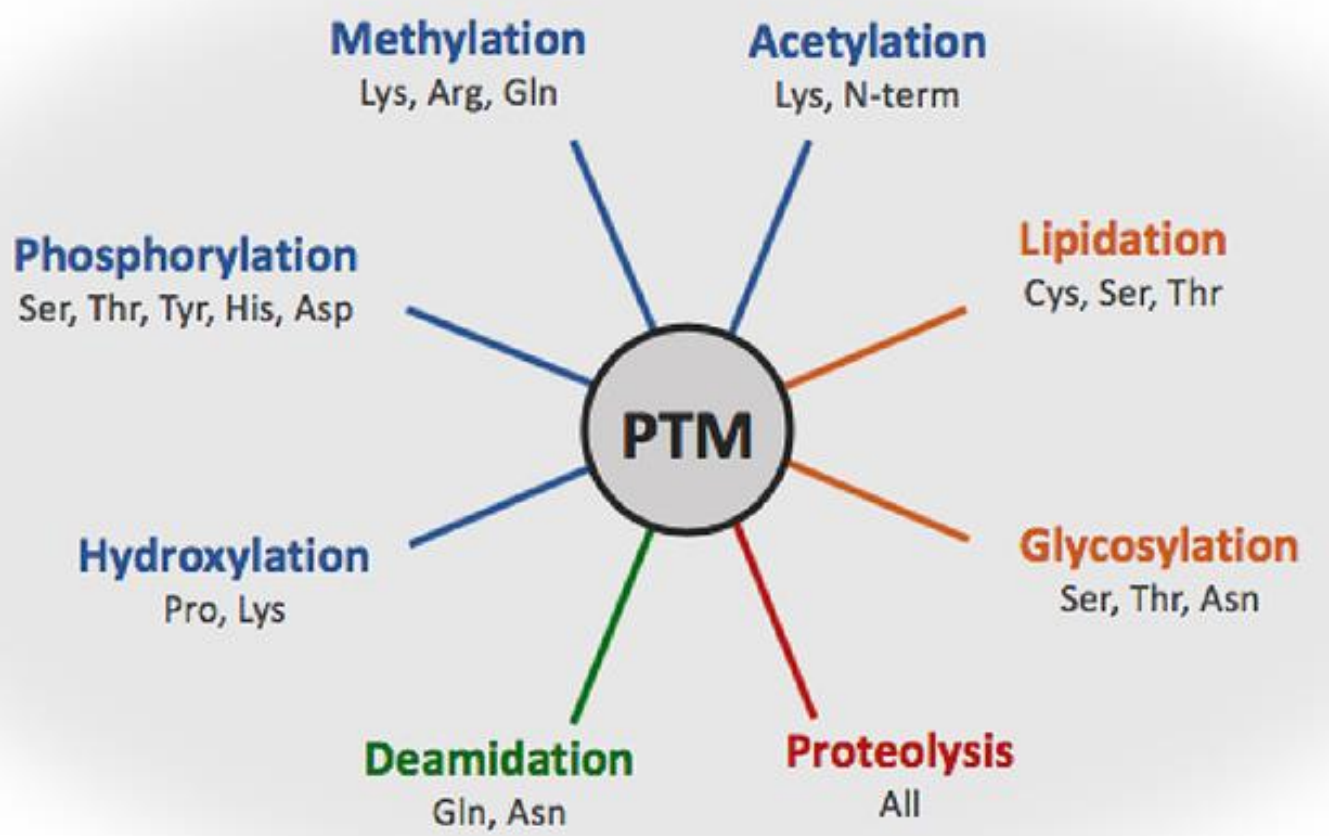
How epigenomes affect our life?

- Epigenome: free radical attached to DNAs of human body.
- Epigenome can facilitate or hinder the expression of useful gene.
- Epigenome can go from generation to generation.
- Epigenomes are highly affected by our lifestyle and our environment.
- You can change your current status of genomes by your habit.

PTM(Post Translational Modification)

- **PTM (Post-Translational Modification)** is when a **protein gets small chemical changes after it's made** to help it **work properly**.
- **Why:** Turn it on/off, fold correctly, go to the right place, or get destroyed.
- **Examples:**
  - **Phosphate added** → switch on/off
  - **Sugar added** → helps folding or transport
  - **Ubiquitin added** → mark for breakdown

PTM(Post Translational Modification)



PTM(Post Translational Modification)

- **Protein = Smartphone → PTM = Installing Apps or Updates**
  - The phone exists, but apps or updates let it **do new functions**.
- **Protein = Car → PTM = Adding Fuel, Tires, or GPS**
  - The car is built, but modifications let it **drive properly or go further**.
- **Protein = Plain Toy → PTM = Adding Batteries, Stickers, or Accessories**
  - The toy works better or looks different after adding extras.

Machine Learning In Bioinformatics

- **Machine Learning in Bioinformatics** involves using algorithms to analyze and interpret biological data. It helps scientists make sense of complex biological systems and large datasets.
- **Applications**
  - **Gene Prediction** – Identifying coding regions in DNA sequences.
  - **Protein Structure Prediction** – Predicting 3D shapes of proteins.
  - **Disease Diagnosis** – Detecting diseases like cancer using gene expression data.
  - **Genome Sequencing** – Classifying and annotating genetic information.

Algorithm Steps

- **Input Sequence Representation**
  - Turn Sequence into vector/feature Matrix
- **ML Algorithm and Model Architecture Design**
  - Build Your own Customized Model.
- **ML Model Selection**
  - During Train Validation Phase
- **Data Balancing in training set**
  - A must be imbalance dataset
  - Undersampling/oversampling technique
- **Test With an independent test set**
  - Final Evaluation

Performance metrics

- Performance metrics are quantitative measures used to evaluate how well a model or system performs — especially in machine learning, classification, regression, and other data science tasks.

Classification Metrics

- TP — true positive, model said “YES” and it’s right! ✅
- FP — false positive, model said “YES” but it’s not right! ❌
- TN — true negative, model said “NO” and that’s right! ✅
- FN — false negative, model said “NO” but that’s not right! ❌

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Classification Metrics

- Precision
  - Precision means “**How many of the items predicted as positive are actually positive?**”
  - Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Classification Metrics

- Recall (Sensitivity or True Positive Rate)
  - Recall means "**How many of the actual positives did the model find?**"
  - Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Classification Metrics

- F1-score
  - **F1-score** is the **balance between Precision and Recall**.
    It gives one single number that combines both.
  - Formula:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Classification Metrics

- Accuracy
  - Accuracy means “**How many predictions were correct out of all?**”
  - Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Classification Metrics

- Specificity (True Negative Rate)
  - Specificity means "**How well the model detects actual negatives correctly?**"
  - Formula:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Sample Question 1

You are testing a model that predicts whether a gene is present in a DNA sequence. The following results were observed:

- Out of 100 samples, 80 genes were correctly predicted as genes.
- The model predicted 20 non-genes as genes.
- 10 genes were incorrectly predicted as non-genes.
- 90 non-genes were correctly predicted as non-genes.

- **Using this information, calculate:**
  - **Accuracy**: What percentage of correct predictions did the model make?
  - **Precision**: How accurate is the model when it predicts a gene to be present?
  - **Recall**: How many of the actual genes were correctly identified by the model?
  - **F1 Score**: What is the balance between precision and recall?

Answer

Given:
Total Samples = 100
True Positives (TP) = 80 (The model correctly predicted 80 genes as genes)
False Positives (FP) = 20 (The model incorrectly predicted 20 non-genes as genes)
False Negatives (FN) = 10 (The model incorrectly predicted 10 genes as non-genes)
True Negatives (TN) = 90 (The model correctly predicted 90 non-genes as non-genes)

1. Accuracy
Accuracy is the proportion of correct predictions (both **True Positives (TP)** and **True Negatives (TN)**) out of all predictions (TP + TN + FP + FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Substitute the values:

$$\text{Accuracy} = \frac{80 + 90}{80 + 90 + 20 + 10} = \frac{170}{200} = 0.85$$

Answer: **Accuracy = 0.85** or **85%**

Answer

2. Precision

Precision is the proportion of correctly predicted positive instances (True Positives) out of all predicted positives (True Positives + False Positives).

$$\text{Precision} = \frac{TP}{TP + FP}$$

Substitute the values:

$$\text{Precision} = \frac{80}{80 + 20} = \frac{80}{100} = 0.8$$

Answer: **Precision = 0.8** or **80%**

3. Recall (Sensitivity)

Recall is the proportion of correctly predicted positive instances (True Positives) out of all actual positives (True Positives + False Negatives).

$$\text{Recall} = \frac{TP}{TP + FN}$$

Substitute the values:

$$\text{Recall} = \frac{80}{80 + 10} = \frac{80}{90} = 0.89$$

Answer: **Recall = 0.89** or **89%**

Answer

4. F1 Score

The **F1 Score** is the harmonic mean of precision and recall. It provides a balance between the two.

$$\text{F1 } Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Substitute the values:

$$\text{F1 } Score = 2 \times \frac{0.8 \times 0.89}{0.8 + 0.89} = 2 \times \frac{0.712}{1.69} = \frac{1.424}{1.69} \approx 0.84$$

Answer: **F1 Score = 0.84** or **84%**

5. Specificity, also known as the **True Negative Rate (TNR)**, is the proportion of correctly predicted negative instances (True Negatives) out of all actual negative instances (True Negatives + False Positives).

Formula for Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

True Negatives (TN) = 90

False Positives (FP) = 20

Calculation:

$$\text{Specificity} = \frac{90}{90 + 20} = \frac{90}{110} = 0.8181$$

Sample Question 2

You are testing a model that predicts whether a protein is present in a given sample. The following results were obtained:

		Actual	**Actual**
		Protein Present (True)	**Protein Absent (True)**
Predicted	**Protein Present (Pred)**	120 (TP)	30 (FP)
Predicted	**Protein Absent (Pred)**	10 (FN)	140 (TN)

Using these results, calculate the following:

1. **Accuracy**: What percentage of correct predictions did the model make?
2. **Precision**: How accurate is the model when it predicts the protein to be present?
3. **Recall**: How many of the actual proteins were correctly identified by the model?
4. **F1 Score**: What is the balance between precision and recall?

Sample Question 3

You are testing a model that predicts the presence of a disease based on genetic data. The following confusion matrix shows the results:

Predicted / Actual	Disease Present (True)	Disease Absent (True)
Disease Present (Pred)	75 (TP)	25 (FP)
Disease Absent (Pred)	20 (FN)	80 (TN)

Using these results, calculate the following:

1. **Accuracy**: What is the model's overall accuracy?
2. **Precision**: How many of the predicted diseases were actually correct?
3. **Recall**: How many of the actual diseases were identified by the model?
4. **F1 Score**: What is the harmonic mean of precision and recall?

Sequence Representation

- In **One Hot representation** , we are not imposing any of our knowledge on our **ML Model**.
- Rather, we are allowing ML model to learn **patterns** from sequence representation of given data.

Sequence Representation

- DNA Representation
  - **DNA one hot representation**

i. Monomer (K=1)

A -> 1000

T-> 0100 → **One hot vector**

C-> 0010

G-> 0001

"**AATC**" = [[1000], [1000], [0100], [0001]]

↓

4 x 4 matrix

= [1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1]

↓

16 Size vector

DNA one hot representation

ii. K-mer Representation

- Overlapping
- Non-overlapping

Overlapping:

K=2,3,4

K=2,3 → Mostly Used

3-mer(Trimer) -> AAT

ATC → 4x4x4 →64 Possible Unique Trimers

TTC

AAA → [1,0,0,0,0,0,……..,0] → 64 size vector (one hot)

Each trimer represented by a 64 size one hot vector.

DNA one hot representation

- **K-mer Representation**

Overlapping 3-mer/trimer: “A A T T C G”

If sequence is **N**
Then, **trimer = N-2**



= AAT, ATT, TTC, TCG = 4



256 size vector → 4x64 size matrix

DNA one hot representation

- **K-mer Representation**

Non-Overlapping 3-mer/trimer: “A A T T C G”



= AAT, TCG = 2



128 size vector → 2x64 size matrix

If sequence is **N**
Then, **trimer = N/3**
Keep Integer part Only

DNA one hot representation

- **K-mer Representation**

  2-mer (dimer)-> AA
  AT → 4x4 →16 Possible Unique dimers
  TT

AA → [1,0,0,0,0,0,.....,0] → 16 size vector (one hot)

Each dimer represented by a 16 size one hot vector.

DNA one hot representation

- **K-mer Representation**

Overlapping 2-mer/diimer: “A A T T C G”



= AA,AT,TT,TC,CG = 5



80 size vector → 5x16 size matrix

If sequence is **N**
Then, **dimer = N-1**

DNA one hot representation

- **K-mer Representation**

Non-Overlapping 2-mer/dimer: “A A T T C G”



= AA,TT,CG→3



48 size vector → 3x16 size matrix

If sequence is **N**
Then, **dimer = N/2**
Keep Integer part Only

DNA one hot representation

- **K-mer Representation**

Overlapping 4-mer: “A A T T C G”

= AATT,ATTC,TTCG→3

768 size vector → 3x256 size matrix

If sequence is **N**
Then, **4-mer = N-3**

DNA one hot representation

- **K-mer Representation**

Non-Overlapping 4-mer: “A A T T C G”

= AATT→1

256 size vector → 1x256 size matrix

If sequence is **N**
Then, **4-mer = N/4**

DNA K-mer Frequency Representation

- K=1,2,3,4
- K=1,2,3 most common

- K=1
- % of A, T, C and G in sample sequence.
- Example: “AATC” sequence of string
    - A T C G = total 4 nucleotide in DNA
    - [0.5(2/4) 0.25(1/4) 0.25(1/4) 0(0/4)]-> 4 size vector.

DNA K-mer Frequency Representation

- K=2(consider Overlapping)
- % of AA, AT, AC……. GG in sample sequence.
- Example: “AAATT” → AA->2, AT->1, TT->1
- Vector size 16 for 2mer/Dimer.
- AA AT AC ….. TT …. GG
- [0.5 0.25 0 0.25 0] 16 size vector

DNA K-mer Frequency Representation

- K=3(consider Overlapping)
- % of AAA, AAT, AAC……. GGG in sample sequence.
- Example: “AAAAC” → AAA->2, AAT->0, AAC->1

- AAA AAT AAC …… GGG = total 64 size vector
- [2/3= 0.66 0 1/3= 0.33 0

Physicochemical property (pp) representation

- **Dimer**

- PP → AA AT AC AG TA .....GG (16)



- Twist→ . . . . . .(16 values)
- Free energy→ . . . . . .(16 values)



- 90 properties in total.

Physicochemical property (pp) representation

- **Dimer**
- Example: “AAAT” → AA AA AT



- 270 size vector  3 x 90 size vector.

Physicochemical property (pp) representation

- **Trimer**
- PP→ AAA AAT AAC . . . . . GGG (64 values)
- GC-content→ . . . . (64 values)
- Roll → . . . . (64 values)



- 12 properties in total
- Total: 12 x 64 size matrix

Physicochemical property (pp) representation

- **Trimer**
- Example: “AAAT” →AAA AAT



- 24 size vector ➔ 2 x 12 size matrix

Psc KNC general tool (DNA)

- Pseudo K-tuple nucleotide composition
- Determination for **DNA sample sequence**

- Result = a vector [uses physicochemical properties]

Protein Representation

- One hot representation
  - Monomer one hot representation

  - A→ [1, 0, 0, ...]
  - R→ [0, 1, 0, ...] → 20 size one hot vector
  - N→ [0, 0, 1, ...]

  - “ARRND” → [[1,0, 0,.....], [0,1, 0, 0.....],[. . . . . .], [. . . . . .], [. . . . . .]]
  - 10 size vector → 5 x 20 size matrix.

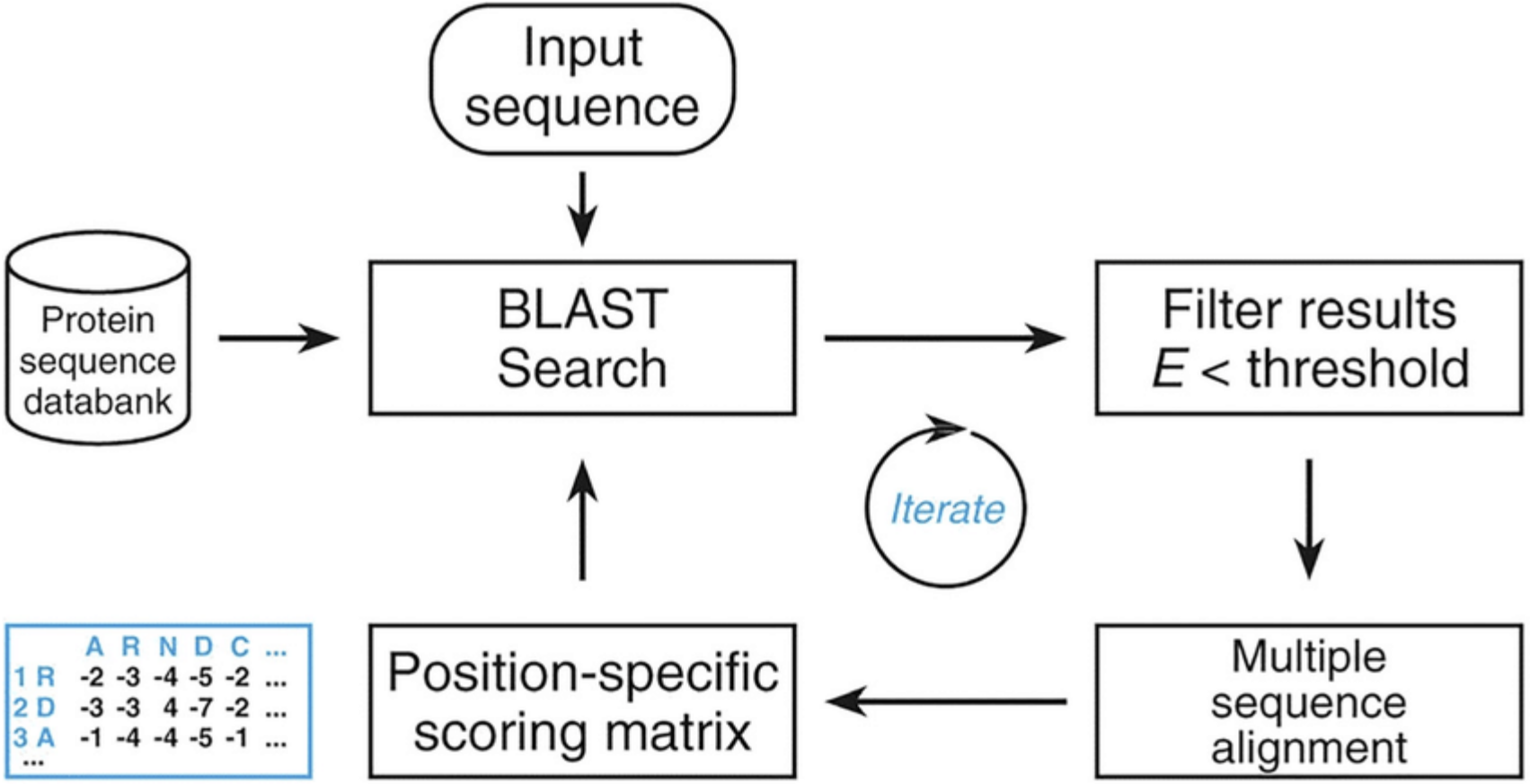
Protein Representation

- PSSM(position specific scoring matrix) representation
- A tool named PSI BLAST (Position Specific Iterative Basic Local Alignment Search Tool) is used.
- Suppose you want to find PSSM for a protein sample.

Protein Representation

- PSI BLAST working steps:
- →takes E-value (threshold value)→ typically 0.0001
- →takes in protein database name.
- → Finds similar proteins which have matching
  - E-val < provided E-val.

PSI BLAST Algorithm



Database Searching Algorithm

- FASTA
  - Query Sequence : **START**
  - Target Sequence: **ARTIST**
  - Value of **K=1**

FASTA

- Step 1: Query Table

1	2	3	4	5
S	T	A	R	T

- Step 2: Hash Table for Query Sequence

A	R	S	T
3	4	1	2 5

FASTA

- Step 3: Extended Target Table

1	2	3	4	5	6
A	R	T	I	S	T
2	2	-1 2	∅	-4	-4 -1

FASTA

- Step 4: Offset Table

-4	-3	-2	-1	0	1	2
2	∅	∅	2	∅	∅	3

- Step 5: Final Table

0	1	2	3	4	5		
S	T	A	R	T			
		A	R	T	I	S	T

FASTA

If 3 found 2 times

- Step 4: Offset Table

-4	-3	-2	-1	0	1	2
2 3	∅	∅	2	∅	∅	3

- Step 5: Final Table 1

0	1	2	3	4	5		
S	T	A	R	T			
		A	R	T	I	S	T

FASTA

Step 5: Final Table 2

-4	-3	-2	-1	0	1	2	3	4
				S	T	A	R	T
A	R	T	I	S	T			

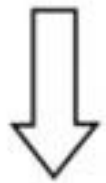
Multiple sequence alignment (MSA)

- **Multiple sequence alignment (MSA)** is a fundamental bioinformatics technique used to align three or more biological sequences (DNA, RNA, or protein) to identify regions of similarity. These alignments provide crucial insights into evolutionary relationships, functional domains, and structural characteristics by highlighting conserved patterns across sequences.

POSITION-SPECIFIC SCORING MATRICES (PSSM)

- a table that contains probability information of amino acids or nucleotides at each position of an ungapped MSA.
- it contains positional information of the alignment
- Rows = residue positions of MSA
- Columns = names of residues
- The values in the table represent log odds scores of the residues calculated from the multiple alignment

Position	1 2 3 4 5 6
Sequence 1	**A T G T C G**
Sequence 2	**A A G A C T**
Sequence 3	**T A C T C A**
Sequence 4	**C G G A G G**
Sequence 5	**A A C C T G**



Convert multiple alignment to a raw frequency table

Pos.	1	2	3	4	5	6	Overall freq.
A	0.6	0.6	—	0.4	—	0.2	0.30
T	0.2	0.2	—	0.4	0.2	0.2	0.20
G	—	0.2	0.6	—	0.2	0.6	0.27
C	0.2	—	0.4	0.2	0.6	—	0.23

Position	1 2 3 4 5 6
Sequence 1	**A T G T C G**
Sequence 2	**A A G A C T**
Sequence 3	**T A C T C A**
Sequence 4	**C G G A G G**
Sequence 5	**A A C C T G**

3/5 = 0.6

Convert multiple alignmen to a raw frequency table

Pos.	1	2	3	4	5	6	Overall freq.
A	0.6	0.6	—	0.4	—	0.2	0.30
T	0.2	0.2	—	0.4	0.2	0.2	0.20
G	—	0.2	0.6	—	0.2	0.6	0.27
C	0.2	—	0.4	0.2	0.6	—	0.23

0.6/ 0.2 = 2.0

Normalize the values by dividing them by overall freq.

Pos.	1	2	3	4	5	6	Overall freq.
A	2.0	2.0	—	1.33	—	0.67	0.30
T	1.0	1.0	—	2.0	1.0	1.0	0.20
G	—	0.74	2.22	—	0.74	2.22	0.27
C	0.87	—	1.74	0.87	2.61	—	0.23

Convert the values to log to base of 2

Pos.	1	2	3	4	5	6
A	1.0	1.0	—	0.41	—	-0.58
T	0.0	0.0	—	1.0	0.0	0.0
G	—	-0.43	1.15	—	-0.43	1.15
C	-0.2	—	0.8	-0.2	1.38	—

POSITION-SPECIFIC SCORING MATRICES (PSSM)

- providing a quantitative description of the degree of sequence conservation at each position of a multiple alignment.
- The **probabilistic model** can then be used like a single sequence for database searching and alignment
- can be used to test how well a particular target sequence fits into the sequence group
- match score = 6.33

$\text{Log}_2(80) \approx 6.33 \Rightarrow 2^{6.33} \approx 80$

- 80 times more likely than by random chance

Match **AACTCG** in the matrix

Find nucleotides at respective pos. of the matrix

Pos.	1	2	3	4	5	6
A	1.0	1.0	—	0.41	—	-0.58
T	0.0	0.0	—	1.0	0.0	0.0
G	—	-0.43	1.15	—	-0.43	1.15
C	-0.2	—	0.8	-0.2	1.38	—

Calculate the sum of log odds scores

1.0 + 1.0 + 0.8 + 1.0 + 1.38 + 1.15 = 6.33

Global Alignment vs. Local Alignment

- Bioinformatics sequence alignment compares DNA, RNA, or protein sequences to identify regions of similarity.

- There are **two main types**:
  - Global Alignment.
  - Local Alignment.

Global Alignment

- Global alignment aligns **two sequences from end to end**, maximizing overall similarity across the entire length.
- **Algorithm**
  - Needleman–Wunsch algorithm
- **When to Use**
  - Sequences are **similar in length**
  - Sequences are **homologous** across their entire span
  - Ideal for **closely related** sequences (e.g., two human genes)

Local Alignment

- Local alignment identifies **the best matching region(s)** between two sequences; it does **not** require end-to-end alignment.
- **Algorithm**
  - Smith–Waterman algorithm
- **When to Use**
  - Sequences differ in length
  - Only a **subregion** is conserved
  - Useful for comparing **distantly related** sequences
  - Ideal for searching **motifs or domains**

Example – Global Alignment

Seq 1: CTAGC

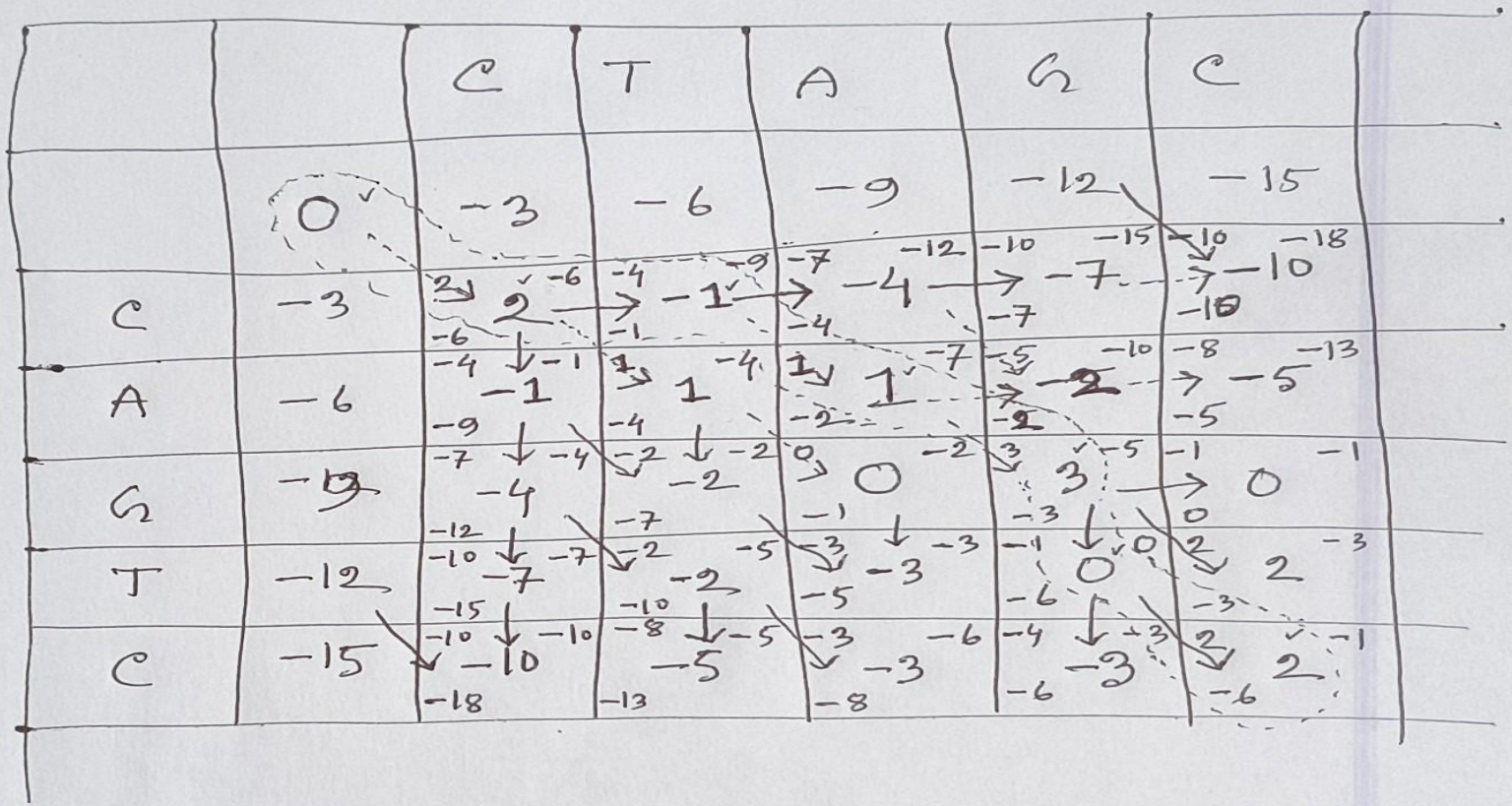
Seq 2: CAGTC

Match Score: +2

Mismatch score: −1

Gap Penalty: −3

1. L/U → Gp +
2. Ma/Mi → v +



Final Alignment:

C T A G _ C

C _ A G T C

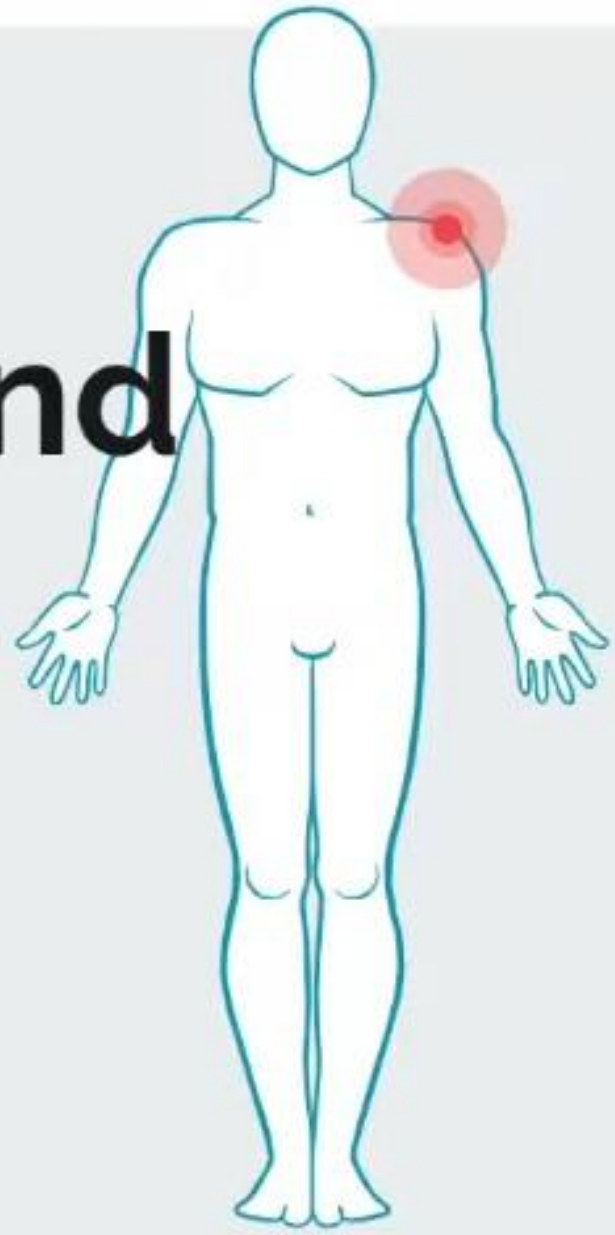
+2, −3 +2 +2 −3 +2 = (+2) → that is the starting value score.

So, Global Alignment is correct.

Local Alignment

- Do this for final exam for same problem??????

Gene Duplication and Read Mapping



CONTENTS

1. Mutation
2. Gene Duplication
3. Read Mapping
   - Keyword Tree
   - Suffix Tree
   - Suffix Array
   - Burrows Wheeler Transform

1. DNA Mutation

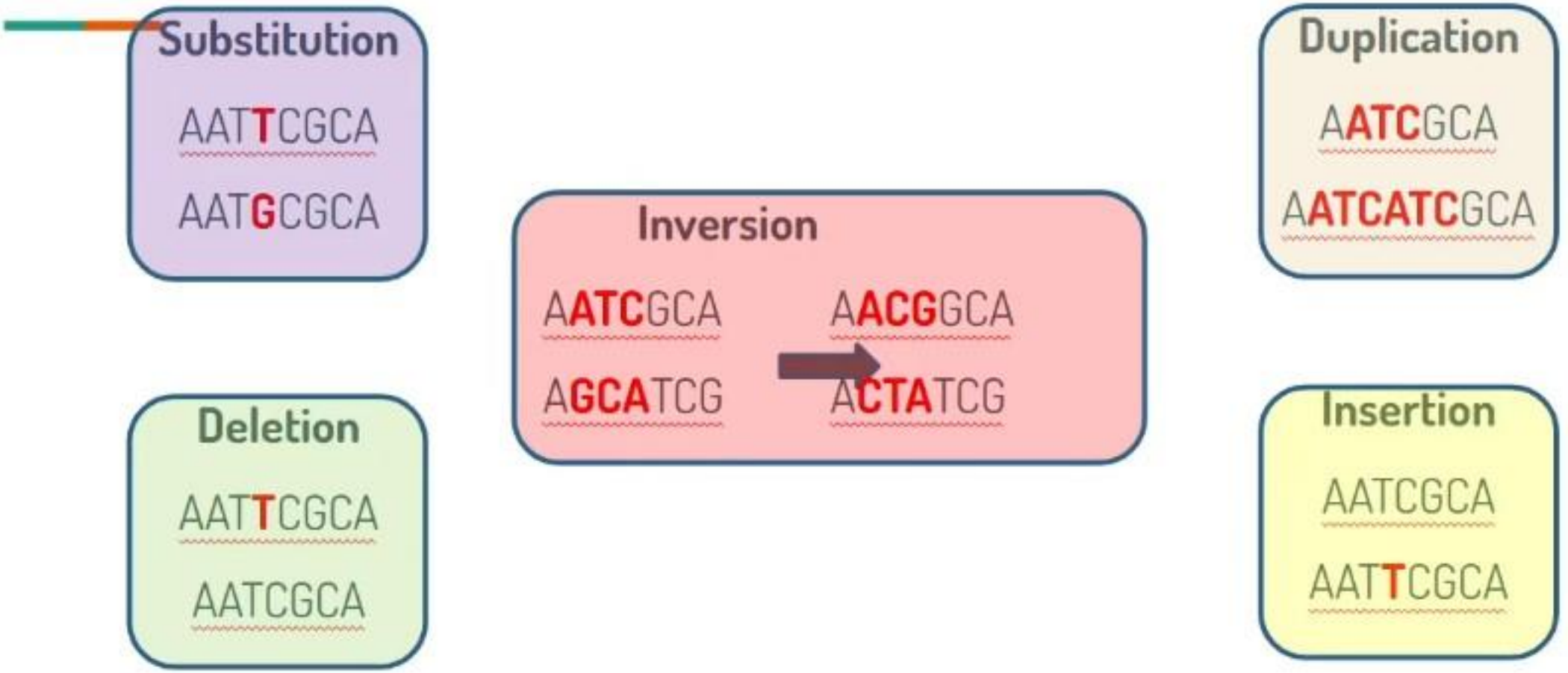
What and how mutation occurs, common forms

Mutation

DNA Mutation refers to sudden, random changes in DNA sequences which leads to different phenotypic expressions.



Common Mutation Types

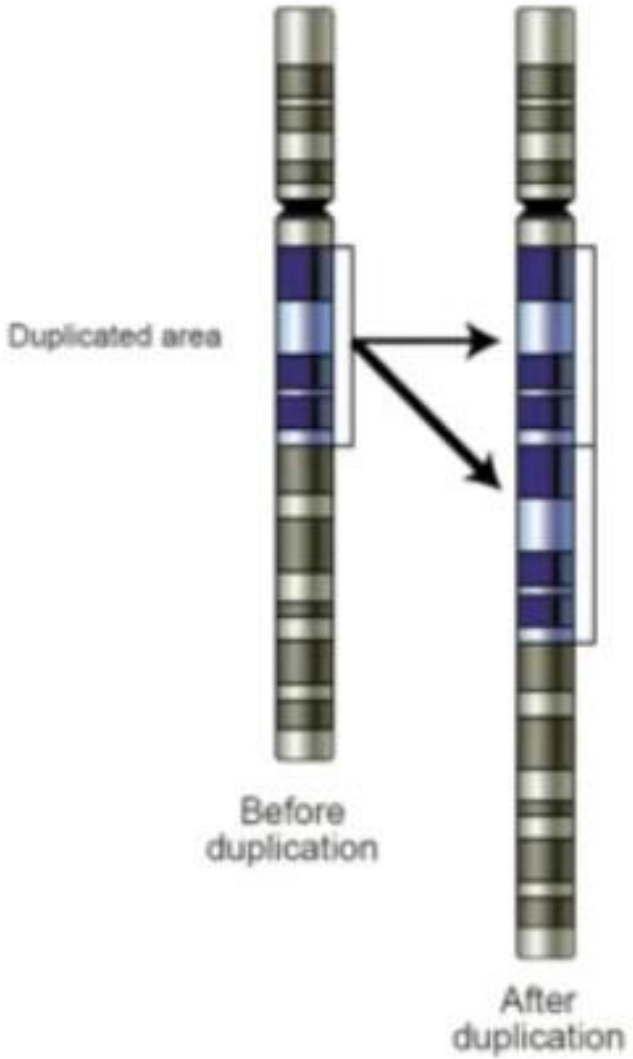


2. Gene Duplication

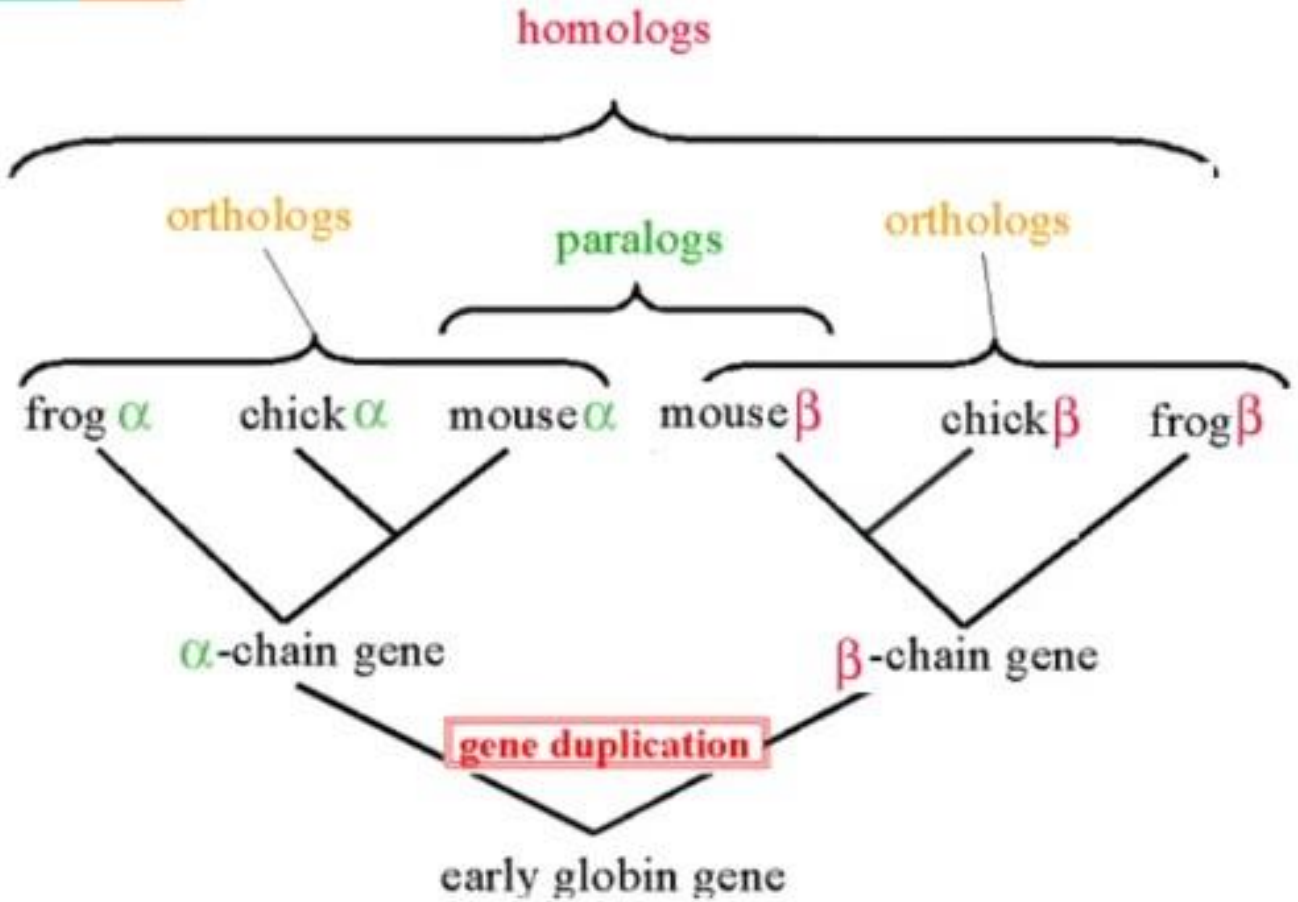
Duplication of Genes, Homolog, Ortholog, Paralogs

Gene Duplication

Gene duplication (or chromosomal duplication or gene amplification) is a major mechanism through which new genetic material is generated during molecular evolution. It can be defined as any duplication of a region of DNA that contains a gene.



Homolog, Ortholog, Paralog and Speciation



- Homolog - A gene related to a second gene by descent from a common ancestral DNA sequence
- Ortholog - Orthologs are genes in different species that evolved from a common ancestral gene by speciation*
- Paralog - Paralogs are genes related by duplication within a genome
- Speciation* - Speciation is the origin of a new species capable of making a living in a new way from the species from which it arose

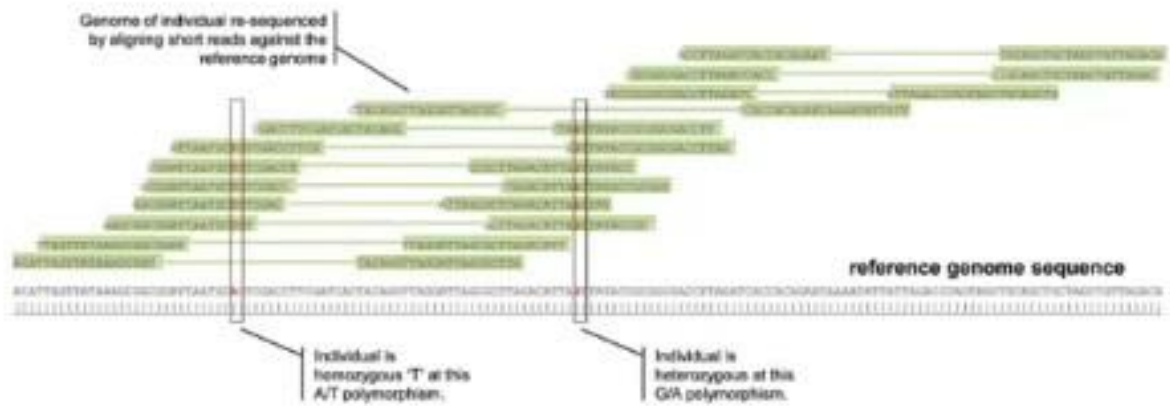
3. Read Mapping

Short Read Mapping, Genome Indexing

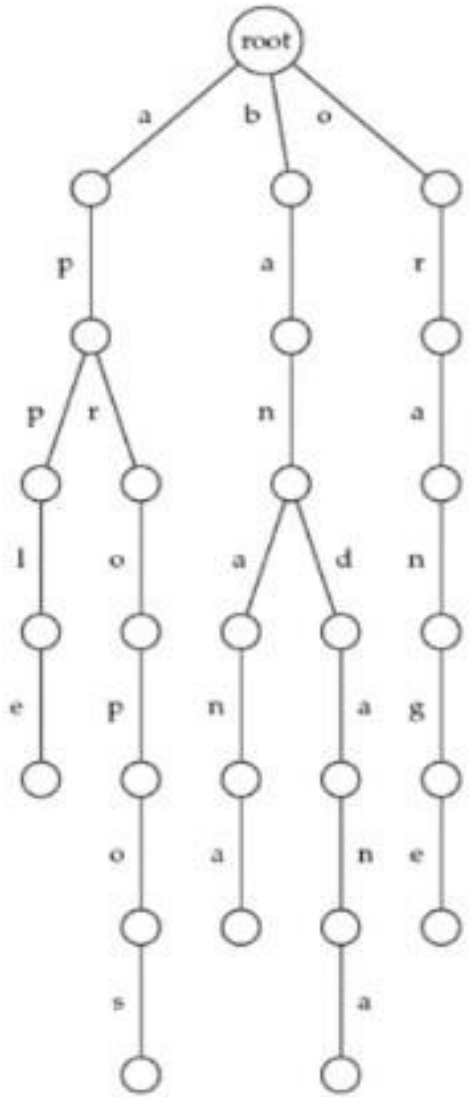
Read Mapping

Mapping refers to the process of aligning short reads to and finding the starting position in a reference sequence (typically Genome).

Short read generally are reads with a length of 30-350 base pairs.



Genome Indexing (Keyword Tree)



- Stores a set of keywords in a rooted labeled tree.
- Each edge is labeled with a letter from an alphabet.
- Any two edges coming out of the same vertex have distinct labels.
- Every keyword stored can be spelled on a path from root to some leaf.
- Furthermore, every path from root to leaf gives a keyword.

Keywords

- Apple
- Apropos
- Banana
- Bandana
- Orange

Genome Indexing

- Genome indexing is a technique used to **efficiently search DNA sequences** inside very large genomes (like the human genome).
  - Suffix Array
  - Suffix Tree
  - Keyword Tree
  - Burrows Wheeler Transform

Genome Indexing

- **Suffix Array:** A **Suffix Array** is a sorted array of all suffixes of a string.
- **Suffix Tree:** A **Suffix Tree** is a compressed tree of all suffixes of a string.
- **Keyword Tree:** A **Keyword Tree** is used to match multiple patterns simultaneously.
- **Burrows Wheeler Transform:** The **Burrows–Wheeler Transform (BWT)** rearranges a string to make it easier to compress and search.

Example

Given the sequence **S = ATTCAT**,
(a) Construct the **Suffix Array**.
(b) Draw the **Suffix Tree**.
(c) Construct the **Keyword Tree (Trie)** for all suffixes.

Answer

- Let’s construct the data structures for the sequence:
  - **S = ATTCAT**
    (Usually, we add a terminal symbol $ to make suffixes unique)
  - S = ATTCAT$

Answer

Step 1: List All Suffixes

Index	Suffix
1	ATTCAT$
2	TTCAT$
3	TCAT$
4	CAT$
5	AT$
6	T$
7	$

Answer

Step 2: Sort Lexicographically

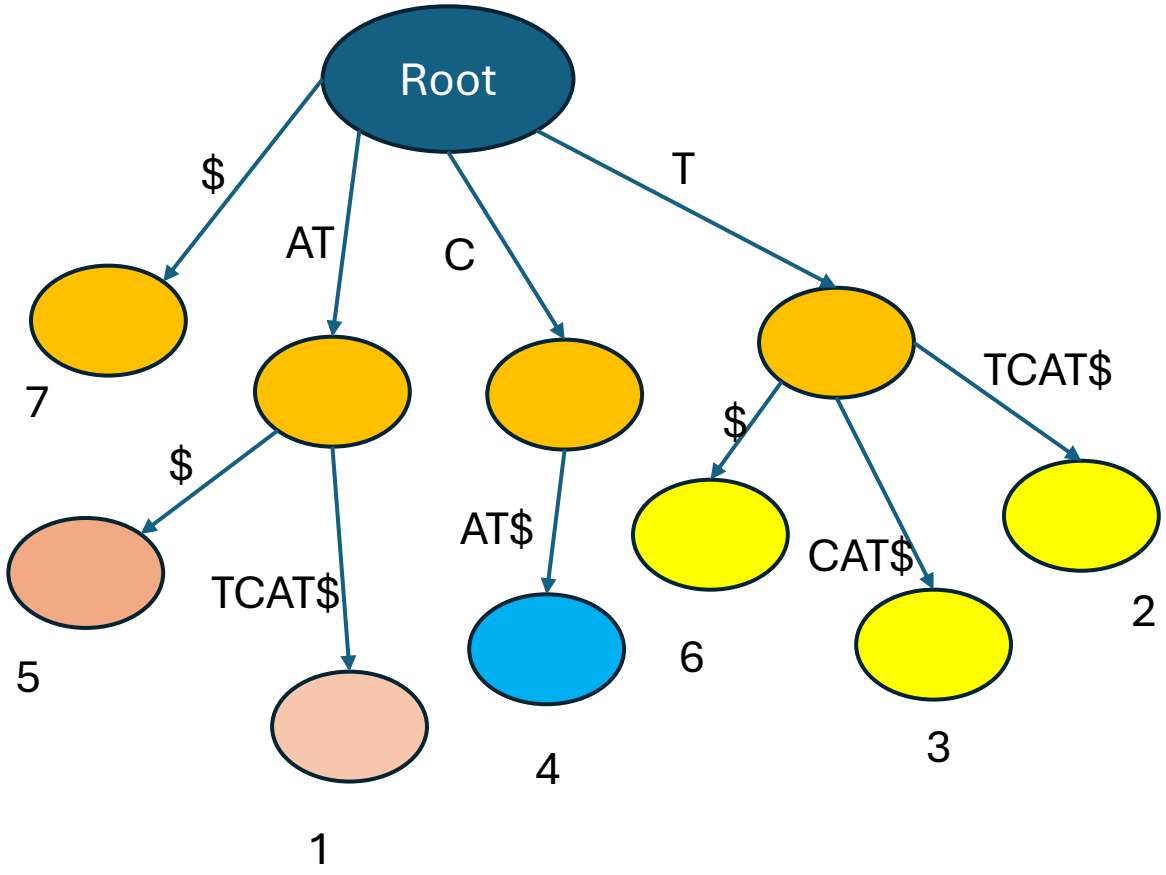
Sorted Order	Suffix	Original Index
1	$	7
2	AT$	5
3	ATTCAT$	1
4	CAT$	4
5	T$	6
6	TCAT$	3
7	TTCAT$	2

Suffix Array:

SA=[7,5,1,4,6,3,2]

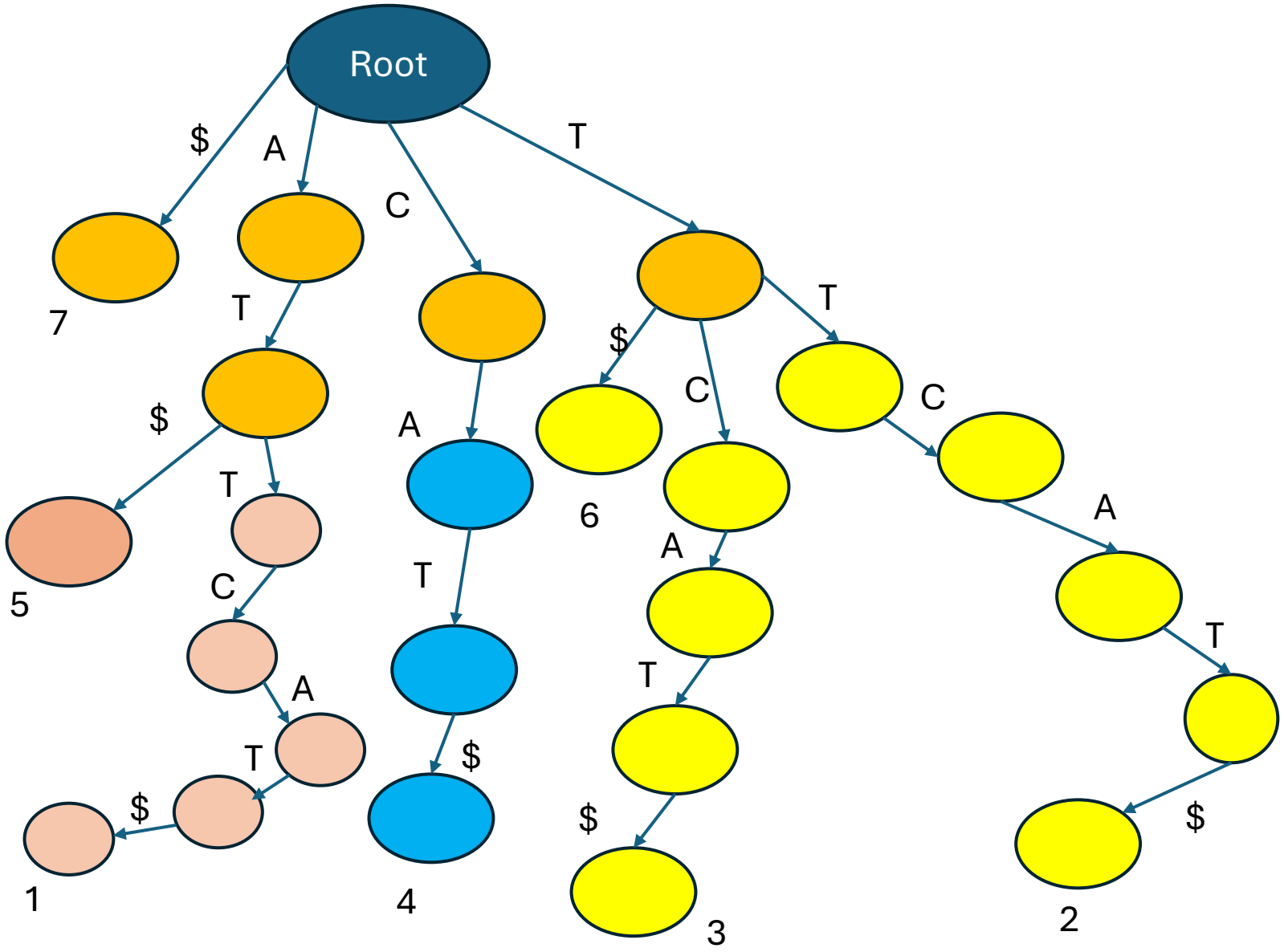
Answer

Suffix Tree:



Answer

Keyword Tree:



Genome Indexing

- Burrows Wheeler Transform

Burrows–Wheeler Transform (BWT) Algorithm

- The Burrows–Wheeler Transform rearranges a string into a form that is easier to compress and search efficiently.
- It does **not compress** the string itself, but prepares it for compression.

BWT Algorithm (Step-by-Step)

Let the string be:

S = ATTCAT

Step 1: Add terminal symbol $

S = ATTCAT$

BWT Algorithm (Step-by-Step)

Step 2: Generate All Cyclic Rotations

Rotation No	Rotation
1	ATTCAT$
2	TTCAT$A
3	TCAT$AT
4	CAT$ATT
5	AT$ATTC
6	T$ATTCA
7	$ATTCAT

BWT Algorithm (Step-by-Step)

Step 3: Sort Rotations Lexicographically

Sorted Order	Rotation
1	$ATTCAT
2	AT$ATTC
3	ATTCAT$
4	CAT$ATT
5	T$ATTCA
6	TCAT$AT
7	TTCAT$A

This is BWM (Burrows Wheeler Matrix)

BWT Algorithm (Step-by-Step)

Step 4: Take Last Column

BWT = TC$TATA

- What is BWM(T)=?

Sorted Order	Rotation
1	$ A T T C A T
2	A T $ A T T C
3	A T T C A T $
4	C A T $ A T T
5	T $ A T T C A
6	T C A T $ A T
7	T T C A T $ A

- What is Suffix Array=?

Sorted Order	Rotation
1	$
2	A T $
3	A T T C A T $
4	C A T $
5	T $
6	T C A T $
7	T T C A T $

- LF Mapping

$	T_0
A_0	C_0
A_1	$
C_0	T_1
T_0	A_0
T_1	T_2
T_2	A_1

LF Mapping:

$ T_0 A_0 C_0 T_1 T_2 A_1

= $ T A C T T A

= A T T C A T $

Molecular Phylogeny
&
UPGMA Implementation

Phylogeny

Phylogeny is the study of evolutionary relationships among organisms, genes, or species.

These relationships are represented using a **phylogenetic tree**.

- **Components of a Phylogenetic Tree**
  - **Root** → Common ancestor
  - **Node** → Divergence point
  - **Branch** → Evolutionary path
  - **Leaf (Taxa)** → Species or gene

What is Molecular Phylogeny?

- Molecular phylogeny reconstructs evolutionary history using:
  - DNA sequences
  - RNA sequences
  - Protein sequences
- Instead of physical characteristics, it uses **sequence similarity** to measure relatedness.

Methods in Molecular Phylogeny

- **Distance-Based Methods**
  - Use pairwise distance matrix
  - Construct tree from similarity/difference
- Examples:
  - **UPGMA**

UPGMA Algorithm

Unweighted Pair Group Method with Arithmetic Mean (UPGMA)

UPGMA is a **hierarchical clustering algorithm** used to construct a rooted phylogenetic tree from a distance matrix.

Example-Type 1

- Given, four sequence (taxas)
  - **A:** A A A A A A A A A A A A A A A A A C G
  - **B:** T T T T T T T A A A A A A A A A A C G
  - **C:** T T T A A A A A T T T A A A A A A C G
  - **D:** A A A T T T A A T T T T T T T T T A C G
- Construct a molecular phylogenetic tree using UPGMA, where

  P-distance = difference/(alignment length)

Answer

1. Construct a distance table using p-distance formula:

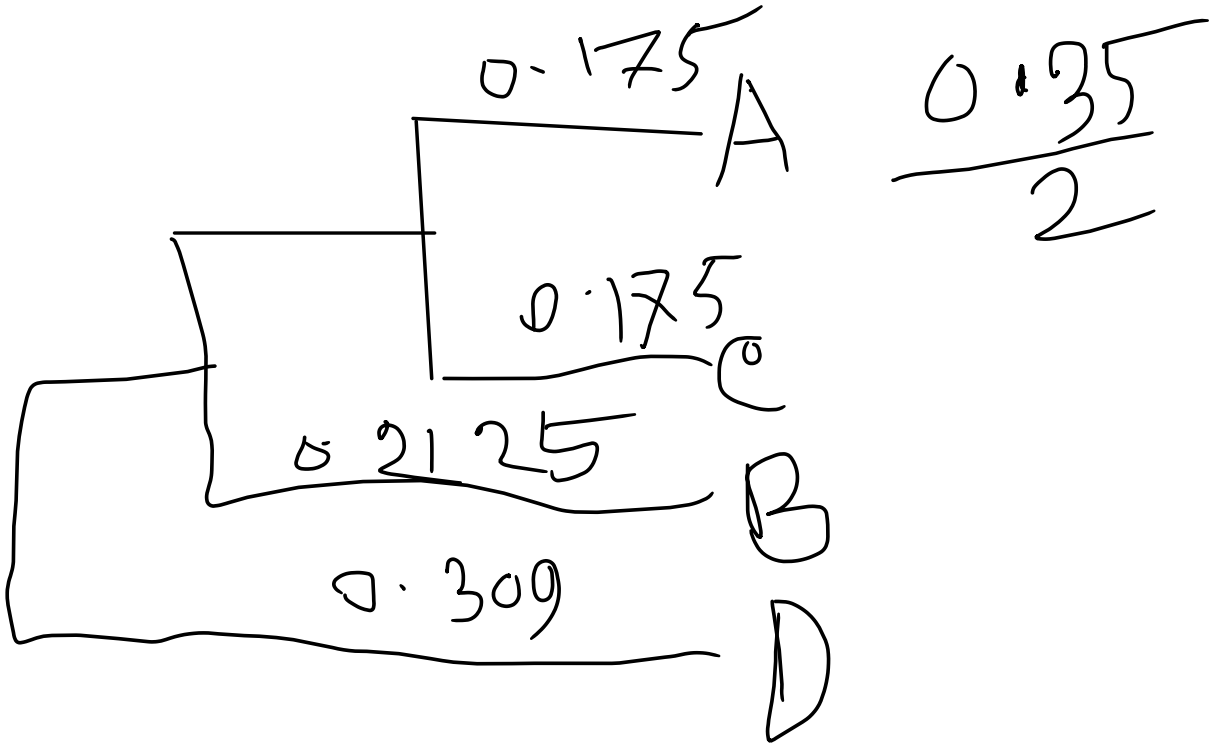
	A	B	C	D
A	0			
B	0.4	0		
C	0.35	0.45	0	
D	0.6	0.7	0.55	0

A vs B
P-distance = 8/20= 0.4
A vs C
P-distance = 7/20 = 0.35

Answer

2. Pick the smallest value, construct the tree, merge the taxas and update the table.

Phylogenetic Tree construction:



Answer

	A-C	B	D
A-C	0		
B	0.425	0	
D	0.575	0.7	0

(AB+BC)/2= (0.4+0.45)/2 = 0.425

(AD+CD)/2= (0.6+0.55)/2 = 0.575

Answer

	B-A-C	D
B-A-C	0	
D	0.617	0

(BD+AD+CD)/3 =
(0.7+0.6+0.55)/3 = 0.617

Example-Type 2

Given, six sequence (taxas):

A:	A	T	C	G	T	G	G	T	A	C	T	G
B:	C	C	G	G	A	G	A	A	C	T	A	G
C:	A	A	C	G	T	G	C	T	A	C	T	G
D:	A	T	G	G	T	G	A	A	A	G	T	G
E:	C	C	G	G	A	A	A	A	C	T	T	G
F:	T	G	G	C	C	C	T	G	T	A	T	C

Construct a molecular phylogenetic tree using UPGMA where,

P-distance = No of difference

Nussinov Algorithm for RNA Secondary Structure Prediction

4. NUSSINOV ALGORITHM (OR BASE PAIR MAXIMIZATION)

- Foundation of RNA structure prediction
- Attempts to maximize RNA base pairs
- Similar to DP algorithms we have already studied
- Does not predict pseudoknots



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BASIC IDEA

- Given an RNA sequence of some length
  - If there are two complimentary bases, score is 1
  - Else, score is 0*
- We find the maximum number of base pairs that are possible in a given sequence

Alignment Method

- Alignment Method
  - Align RNA strand to itself
  - Score increases for feasible base pairs
  - ~~Do not~~ consider G/U for this example

We will

G - C

A - U

G - U Wobble

- Each score independent of overall structure

- Bifurcation adds extra dimension (don't worry about this)

	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}
A_1													
A_2													
U_3													
C_4													
U_5													
G_6													
U_7													
U_8													
A_9													
C_{10}													
G_{11}													
C_{12}													
A_{13}													

	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}
A_1	0												
A_2	0	0											
U_3	0	0	0										
C_4	0	0	0	0									
U_5	0	0	0	0	0								
G_6	0	0	0	0	0	0							
U_7	0	0	0	0	0	0	0						
U_8	0	0	0	0	0	0	0	0					
A_9	0	0	0	0	0	0	0	0	0				
C_{10}	0	0	0	0	0	0	0	0	0	0			
G_{11}	0	0	0	0	0	0	0	0	0	0	0		
C_{12}	0	0	0	0	0	0	0	0	0	0	0	0	
A_{13}	0	0	0	0	0	0	0	0	0	0	0	0	0

	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}
→ A_1	0	0	1	1	2	2	2	3	3	3	3	3	4
→ A_2	0	0	1	1	1	1	2	2	2	2	3	3	4
– U_3	0	0	0	0	0	1	1	1	2	2	3	3	4
C_4	0	0	0	0	0	1	1	1	2	2	3	3	4
U_5	0	0	0	0	0	1	1	1	2	2	3	3	4
G_6	0	0	0	0	0	0	1	1	1	2	2	3	3
U_7	0	0	0	0	0	0	0	0	1	1	2	2	2
U_8	0	0	0	0	0	0	0	0	1	1	1	1	2
A_9	0	0	0	0	0	0	0	0	0	0	1	1	1
C_{10}	0	0	0	0	0	0	0	0	0	0	1	1	1
G_{11}	0	0	0	0	0	0	0	0	0	0	0	1	1
C_{12}	0	0	0	0	0	0	0	0	0	0	0	0	0
A_{13}	0	0	0	0	0	0	0	0	0	0	0	0	0

	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}
A_1	0	0	1	1	2	2	2	3	3	3	3	3	4
A_2	0	0	1	1	1	1	2	2	2	2	3	3	4
U_3	0	0	0	0	0	1	1	1	2	2	3	3	4
C_4	0	0	0	0	0	1	1	1	2	2	3	3	4
U_5	0	0	0	0	0	1	1	1	2	2	3	3	4
G_6	0	0	0	0	0	0	1	1	1	2	2	3	3
U_7	0	0	0	0	0	0	0	0	1	1	2	2	2
U_8	0	0	0	0	0	0	0	0	1	1	1	1	2
A_9	0	0	0	0	0	0	0	0	0	0	1	1	1
C_{10}	0	0	0	0	0	0	0	0	0	0	1	1	1
G_{11}	0	0	0	0	0	0	0	0	0	0	0	1	1
C_{12}	0	0	0	0	0	0	0	0	0	0	0	0	0
A_{13}	0	0	0	0	0	0	0	0	0	0	0	0	0

5′	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}	3′
→ A_1	0	0	1	1	2	2	2	3	3	3	3	3	4	
→ A_2	0	0	1	1	1	1	2	2	2	2	3	3	4	
– U_3	0	0	0	0	0	1	1	1	2	2	3	3	4	
C_4	0	0	0	0	0	1	1	1	2	2	3	3	4	
U_5	0	0	0	0	0	1	1	1	2	2	3	3	4	
G_6	0	0	0	0	0	0	1	1	1	2	2	3	3	
U_7	0	0	0	0	0	0	0	0	1	1	2	2	2	
U_8	0	0	0	0	0	0	0	0	1	1	1	1	2	
A_9	0	0	0	0	0	0	0	0	0	0	1	1	1	5′
C_{10}	0	0	0	0	0	0	0	0	0	0	1	1	1	
G_{11}	0	0	0	0	0	0	0	0	0	0	0	1	1	
C_{12}	0	0	0	0	0	0	0	0	0	0	0	0	0	
A_{13}	0	0	0	0	0	0	0	0	0	0	0	0	0	

U
U_7 — A_9
G_6 — C_{10}
U_5 — G_{11}
C_4 C_{12}
U_3 — A_{13}
A
A
A
3′
5′

Thank You

Any Question?